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Simulation for Detection Efficiency Enhancement at the Ultracold YbF Electron Electric Dipole Moment Experiment

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Abstract

The electron electric dipole moment (eEDM) is a powerful probe of additional CP violation and theories beyond the Standard Model. Modern experiments use heavy polar molecules like YbF whose strong effective electric fields enhance the extremely small energy shift the theoretical eEDM would produce. Measuring the eEDM hence requires unprecedented precision when reading out quantum state populations. The ultracold eEDM experiment at Imperial College London utilises a rotationally closed optical cycling transition, cycling the molecules to near completion. However, the statistical sensitivity is limited by the fraction of molecules that go dark. On average, about 13.4 photons are scattered per molecule before being lost to higher vibrational states where they do not contribute to the signal anymore. A new detection scheme has been implemented and tested where a V1 re-pump laser was added to recover these lost molecules and return them to the cycling transition. To assess its effectiveness, a computational monte-carlo wavefunction (MCWF) simulation based on the Lindblad equation is developed and validated against experimental data. The simulation is accurate across a wide range of parameters at low laser power but underestimates the scattering ratio at high power, most likely due to additional dark state re-mixing channels in the experiment, among them spatial polarisation gradients. The new cycling scheme is found to increase the average number of photons scattered per molecule to 25.6 at moderate V1 power of 27 mW. The simulation framework was then used to examine the potential of magnetic fields and polarisation modulation to maximise the scattering ratio further. It is concluded that an externally applied magnetic field between 0.5 and 1 G improves the re-pump laser enhancement, whereas raising the polarisation modulation frequency yields only marginal gains. Using the MCWF, it is estimated that the new detection scheme can increase sensitivity by a factor of at least 1.3, though its feasibility relies on the ability to clean-up the newly incurred ground-state populations it leaves behind.

Declaration of Originality

I declare that the work presented in this thesis is my own. Where required, I have acknowledged the work of others and made reference to the correct sources.

This project was conducted in collaboration with Zak Farley. Some of the code and figures produced are his own. Where they have been included in the thesis, permission to do so has been given.

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1

Introduction

1.1 Overview of the subject matter

1.1.1 Motivation

One of the most significant unresolved problems in modern physics concerns the baryon asymmetry; the question of why the universe contains more matter than anti-matter. Adndrey Sakharov identified three conditions for the baryon asymmetry to form cosmologically [1]:

1. Violation of of baryonic charge conservation,
2. a difference between particles and antiparticles, manifesting itself as CP-invariance violation,
3. and a departure from local thermodynamic equilibrium since non-stationarity formation of baryon asymmetry is only possible under non-stationary conditions.

The CP-violation in the Standard Model (SM), arising from the CKM-matrix together with the PMNS phase are orders of magnitude too small to account for the observed asymmetry [2]. The electron electric dipole moment (eEDM) could give promising evidence for the existence of this so far missing CP violation.

The eEDM is an intrinsic property of the electron determined by its charge distribution

$$d_e = \int_V \rho(\boldsymbol{r}')(\boldsymbol{r}' - \boldsymbol{r}_0)\ d^3\boldsymbol{r}' \tag{1.1}$$

where $\boldsymbol{r}_0$ is centre of mass of the electron in its rest frame. If d_E is non-zero, its interaction with the electric field $\boldsymbol{\varepsilon}$ produces an energy shift given by

$$H_e = -d_e \boldsymbol{\sigma} \cdot \boldsymbol{\varepsilon} \tag{1.2}$$

where σ are the Pauli spin matrices. By the Wigner-Eckhart theorem, the eEDM must be permanently aligned along the electron's spin [3]. Under a parity operation, a polar vector such as $\boldsymbol{\varepsilon}$ transforms as $\boldsymbol{\varepsilon} \to -\boldsymbol{\varepsilon}$ whereas an axial vector like $\boldsymbol{\sigma}$ is unchanged. Under time reversal, $\boldsymbol{\varepsilon}$ is left invariant while $\boldsymbol{\sigma} \to -\boldsymbol{\sigma}$ as the direction of rotation is reversed. This is illustrated in figure 1.1. It follows immediately from equation 1.2, that eEDM violates both P and T. The CPT theorem states that any quantum field theory obeying Lorentz invariance must conserve CPT [4], so T violation must be equivalent CP violation. A measurement of d_e would therefore be evidence for additional CP violation beyond the SM and would directly probe Sakharov's second condition. It would also make the eEDM a sensitive probe

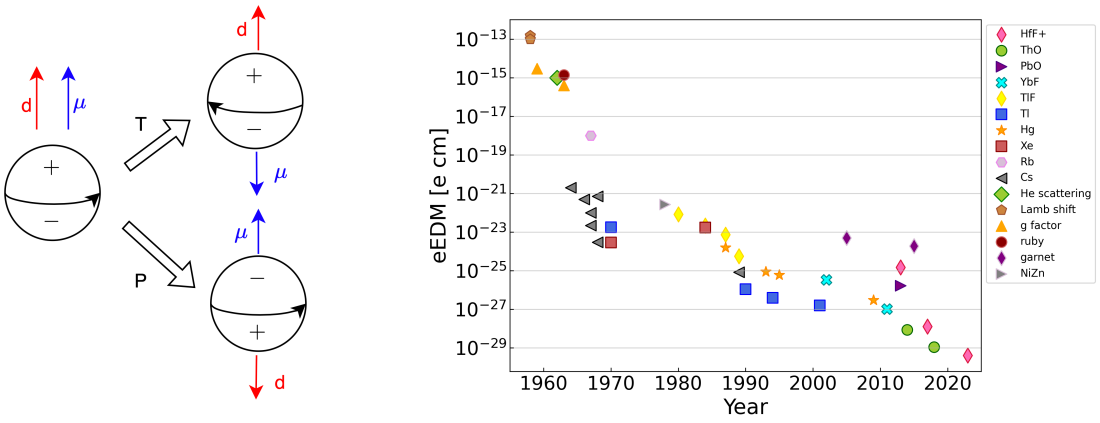


Figure 1.1: **Left**: P and T violation of the eEDM. d denotes the eEDM and μ is the magnetic moment. **Right**: eEDM sensitivity over time [9].

for beyond the standard model (BSM) theories. Particularly, any value larger than the SM prediction [5]

$$|d_e| < 10^{-38}\text{e cm}, \tag{1.3}$$

which lies far below current experimental limits, would be irrefutable evidence of BSM physics. The SM value is so small because contributions to d_e from the lepton sector are negligible [6], and higher order radiative corrections involving virtual quarks via $W^{\pm}$-bosons either only contribute a CP-violating phase or first enter at 4th order [5, 7]. Many BSM theories, by contrast, admit new CP-violating scalar fields that generate d_e at lower orders and hence at larger magnitude. Supersymmetry (SUSY), for example, aims to solve the hierarchy problem by introducing new supersymmetric partners for each SM particle and predicts $\sim 10^{-28}$e cm [8]. Current experimental limits have already largely this parameter space prompting further theoretical work on SUSY. Steadily improving eEDM experiments therefore continuously inform and constrain leading BSM theories.

1.1.2 Research Environment

A wealth of experiments has been developed to measure the eEDM, continuously pushing frontier limits and placing ever tighter constraints on BSM theories. The improvement in sensitivity over time is shown in the right in figure 1.1. Much of this progress has come from both improving experimental conditions and the choice of molecule. Most experiments use heavy molecules whose large effective electric fields enhance the eEDM effects [10].

The most sensitive result to date comes from the JILA collaboration which reported $|d_e|<4.1\times10^{-30}$e ċm at 90% confidence in 2022 [11]. The experiment utilises 20,000 HfF^+ molecules held in a radio-frequency ion trap which allows for very long coherence and measurement times. The internal effective electric field $\varepsilon_{\text{eff}} = 23$ GVcm^{-1} acts on the electron's spin to enhance the eEDM signal [5]. A coherent superposition of the $|v = 0, J = 1, m_F = \pm\frac{3}{2}\rangle$ spin states is prepared and the energy splitting, which contains the eEDM contribution $\pm 2 d_e \varepsilon_{eff}$ is measured using Ramsey spectroscopy [11]. Trapping times of ~ 3 s allow the experiment to reach very low uncertainties.

ACME II takes a different approach, passing ThO molecules through a 20cm interaction region. ThO has a larger effective electric field $\varepsilon_{\text{eff}} = 78$ GVcm^{-1} [12] than HfF^+ and the interaction region allows for a larger number of molecules at the cost of a coherence time of only 1 ms. The result is the second lowest bound to date of $|d_e| < 1.1 \times 10^{-29}$e ċm [12].

The experiment considered in this work was developed by Imperial College London and reported an upper bound of $|d_e| < 1.06 \times 10^{-27}$e cm in 2011 [13] with potential to reach a statistical uncertainty below $10^{-30}e$ cm [14]. It was first molecular measurement of its kind.

1.2 Project Outline

One key limitation in eEDM sensitivity is the statistical sensitivity of the detection region and its signal to noise ratio. Dark states, residual populations, and noise make this difficult to optimise in practice. Simulations offer valuable ways to assess how various experimental parameters affect detection sensitivity and noise statistics. In this project, a comprehensive Monte Carlo wavefunction (MCWF) simulation was developed to better understand the new detection scheme. The relevant detector components are described in chapter 2 and the code is described in chapter 3. Finally, in chapter 4, the simulation is compared to experimental data, detection efficiency improvements are evaluated, and simulation limitations are discussed.

2

The ultracold eEDM experiment

2.1 Molecular Energy Level Structure of YbF

2.1.1 Choice of YbF

Theoretically, the eEDM would manifest itself as a shift in the energy levels of a particle placed in an electric field. Schiff's theorem [15] states that in a neutral molecule such as YbF, there is no first-order shift in the interaction energy due to the eEDM. The internal charge distribution rearranges itself so that the average field at the electron vanishes, provided the dynamics are point-like and non-relativistic. Sandars [16], however, showed that second- and third-order relativistic effects enhance the observable eEDM energy shift. In heavy molecules, the coulomb potential is much larger and electron speeds are on average higher making relativistic effects more pronounced. This can be thought of as the electron experiencing a higher effective field ε_{eff}. The ultracold experiment uses a molecular beam of YbF, which when fully polarised yields $\varepsilon_{\text{eff}} = 25$ GVcm^{-1} [17]. This effective field enhances the measured eEDM shift

$$\Delta E_{d_e} = d_e \eta(\varepsilon_{\text{ext}}) \varepsilon_{\text{eff}} \tag{2.1}$$

where $\eta(\varepsilon_{\text{ext}})$ quantifies the degree of molecular alignment in the externally applied field ε_{ext}.

This section summarises the energy level structure of YbF and fixes notation. A full treatment of the effective Hamiltonian is given by Brown and Carrington [18] and a more comprehensive treatment specific to YbF molecules by Fitch and Tarbutt [19].

2.1.2 Born-Oppenheimer and Other Approximations

Because the nuclei are far heaver than the electrons, the electric motion is much faster and follows adiabatically. YbF is therefore well described by the Born-Oppenheimer, rigid-rotor, and harmonic-oscillator approximations [20]. This allows the nuclear and electronic wavefunction to be decoupled and a separate set of quantum numbers is defined for each. The quantum numbers are defined in table 2.1. Throughout, we work in cylindrical polar coordinates with $\hat{z}$ along the axis connecting the two nuclei, called the internuclear axis.

The electronic states are described using standard molecular notation

$$\rho^{2S+1}\Lambda_{\Omega}^{(\pm)} \tag{2.2}$$

where ρ is the state identifier with X for the ground state and A for the first excited state. Λ and Ω are the projections of L and J onto the internuclear axis as detailed in table 2.1. By convention, the greek symbols $\Sigma, \Pi, \Delta, ...$ are used for $\Lambda = 0, 1, 2, ...$ The superscript $\pm$ denotes the reflection symmetry of the state through a plane orthogonal to the internuclear axis. With a single valence electron in YbF, $S = 1/2$.

The order of coupling of the quantum numbers are given by Hund's cases for diatomic molecules where some interactions are assumed to be dominating. Quantum numbers are then coupled with the strongest couplings first [21].

Quantum Number	Definition	Hund's case
L	Electron orbital angular momentum	a & b
S	Electron spin	a & b
R	Rotational angular momentum of nuclei axis	a & b
I	Nuclear spin	a & b
$J = R + L + S$	Angular momentum excluding nuclear spin	a
$F = J + I$	Total angular momentum	a
$N = L + R$	Angular momentum excluding spins	b
$G = I + S$	Total spin angular momentum	b
$F = G + N$	Total angular momentum	b
Λ	Projection of L onto internuclear axis	a & b
Σ	Projection of S onto internuclear axis	a
Ω	Projection of J onto internuclear axis, $\Lambda + \Sigma$	a

Table 2.1: Quantum Numbers fully describing YbF

2.1.3 Excited state

The excited state is best described by Hund's case (a), in which the electrostatic interaction dominates, followed by the spin-orbit interaction, while rotational coupling is negligibly weak. This means that $\boldsymbol{L}$ is electro-statically coupled to the internuclear axis precessing fast around it and $\boldsymbol{S}$ is in turn coupled to $\boldsymbol{L}$ through the spin-orbit interaction. Both $\boldsymbol{L}$ and $\boldsymbol{S}$ have projections along the internuclear axis $\hat{z}$ denoted Λ and Σ respectively. We may define the vector $\boldsymbol{\Omega} = \boldsymbol{\Lambda} + \boldsymbol{\Sigma}$ which points along $\hat{z}$ and is a good quantum number. Because the rotational angular momentum $\boldsymbol{R}$ is always perpendicular to the internuclear axis by definition, it follows that $\boldsymbol{J} = \boldsymbol{R} + \boldsymbol{\Omega}$ is itself a good quantum number. The good quantum numbers are Λ, S, Σ, J, and Ω. A state of definite Ω, which can be aligned $|+\Omega\rangle$ and anti-aligned $|-\Omega\rangle$, is not a parity eigenstate. Energy levels require definite parity and we therefore form

$$\begin{aligned} |e\rangle &= \frac{1}{\sqrt{2}}\left(|+\Omega\rangle + |-\Omega\rangle\right) \\ |f\rangle &= \frac{1}{\sqrt{2}}\left(|+\Omega\rangle - |-\Omega\rangle\right) \end{aligned} \tag{2.3}$$

which have parity $P = \pm(-1)^{J-1/2}$. Finally, the nuclear spin is coupled $\boldsymbol{F} = \boldsymbol{J} + \boldsymbol{I}$. Since $I = 1/2$ for YbF, the $J = 1/2$ state admits the hyperfine levels $F = 1$ and $F = 0$. The hyperfine structure of the excited state $A^2\Pi_{1/2}$ is shown in figure 2.1.

2.1.4 Ground State

The ground state is well described by Hund's case (b), in which spin-orbit coupling is negligible and rotational coupling is comparatively strong, so that $\boldsymbol{S}$ couples to $\boldsymbol{R}$ rather than $\boldsymbol{L}$. The dominant electrostatic interaction again causes $\boldsymbol{L}$ to precess quickly, so Λ remains a good quantum number. However, Σ and Ω are no longer defined. Because the nuclear-spin interaction is stronger than the spin-rotation interaction, $\boldsymbol{I}$ is coupled to $\boldsymbol{S}$ first, via $\boldsymbol{G} = \boldsymbol{I} + \boldsymbol{S}$. Since $S = I = 1/2$, it follows that $G = 1, 0$. The ground state has $L = 0$ so that $N = R = 0, 1, 2, ...$ with parity $P = (-1)^N$. The hyperfine levels then follow from $\boldsymbol{F} = \boldsymbol{G} + \boldsymbol{N}$.

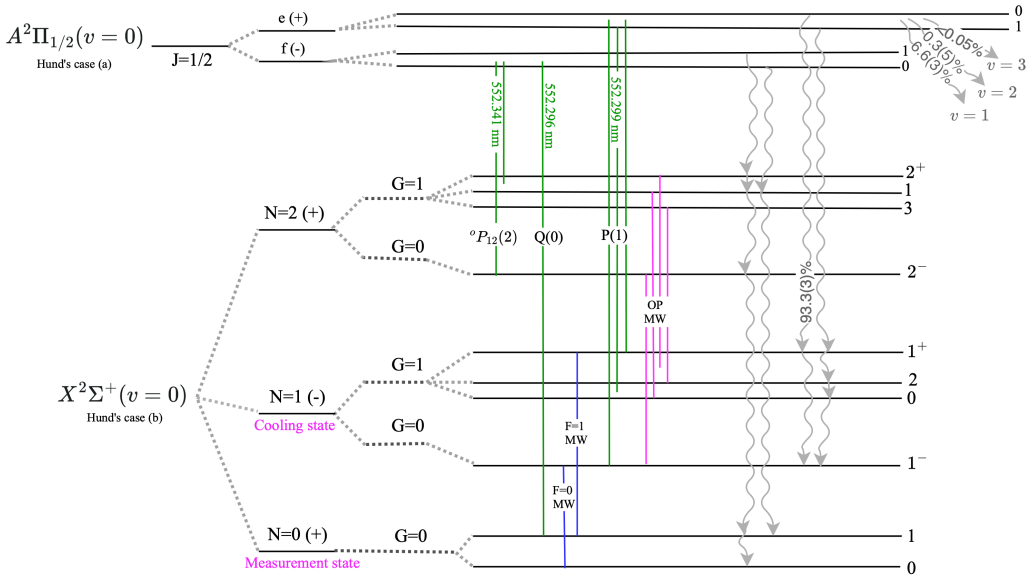


Figure 2.1: Hyperfine level structure of the Ground and first excited state of YbF for $v = 0$. Laser transitions are in green, microwave transitions are in blue and pink. The $(+)$ and $(-)$ next to states denotes their parity. The $+/-$ as a superscript denotes higher/lower energy state for the same F. For a comprehensive view with all rotational and angular momentum levels, see appendix A

Together, the Hund's cases (a) and (b) account for all relevant YbF energy levels relevant here. The most frequently addressed ones are shown in figure 2.1.

2.2 Electromagnetic Interactions

Electric and magnetic fields further split the hyperfine levels established above through the Stark effect, the Zeeman interaction, and the eEDM interaction.

2.2.1 Stark Effect

The molecule possesses an electric dipole moment $\mu_E = 1.97$ MHz V^{-1}cm [22] directed along the internuclear axis. In an electric field,

$$H_S = -\boldsymbol{\mu}_E \cdot \boldsymbol{\varepsilon}_{\text{ext}} = -\mu_E \varepsilon_{\text{ext}} \cos(\theta). \tag{2.4}$$

Since $\cos(\theta)$ is parity odd, it mixes states with opposite parity, i.e. $\Delta N = \pm 1$ in $X^2\Sigma^+$ which are energetically close together. This rotational level structure causes the mixing to be substantial and the energy shift is large.

The $|F = 1, m_F = 0, \pm 1\rangle$ states are relevant for the eEDM measurement. The Stark effect moves the $m_F = \pm 1$ level up and the $m_F = 0$ level down causing a split of about 15MHz at $\varepsilon_{\text{ext}} = 20$kV cm^{-1}.

2.2.2 Zeeman Interaction

The magnetic field interaction is given by

$$H_Z = g_S \mu_B \boldsymbol{S} \cdot \boldsymbol{B} \tag{2.5}$$

which yields an energy shift

$$\Delta E_Z = g_F \mu_B B m_F. \tag{2.6}$$

The Zeeman interaction coupling the nuclear spin $\boldsymbol{I}$ with $\boldsymbol{B}$ has been neglected as it is much smaller than equation 2.6.

2.2.3 eEDM Interaction

The eEDM interaction is a stark shift due to the permanent d_e which interacts with the effective electric field pointing along the internuclear axis. By the Wigner-Eckhart theorem, the eEDM must be permanently aligned with the electron spin. Then,

$$H_{\text{eEDM}} = -\boldsymbol{d}_e \cdot \boldsymbol{\varepsilon}_{\text{eff}} = d_e |\boldsymbol{S}| \eta \varepsilon_{\text{eff}}. \tag{2.7}$$

The splitting of the hyperfine structure in $|N = 0\rangle$ due to the Stark effect, Zeeman interaction, and eEDM interaction is shown in figure 2.2.

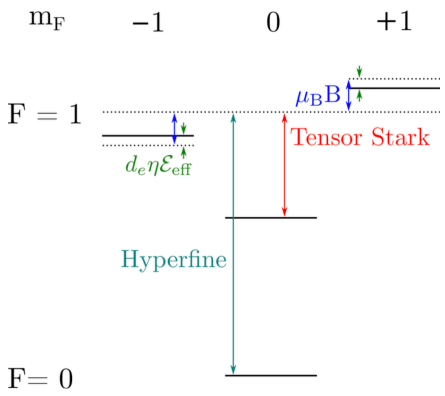


Figure 2.2: Hyperfine energy level structure of $|N = 0\rangle$. The splitting is not to scale. Figure from [23].

2.3 Experimental Setup

The experimental setup of the ultracold eEDM experiment is shown in figure 2.3. Each component is discussed in turn below, with particular emphasis on the detection region and parts which have been computationally modelled.

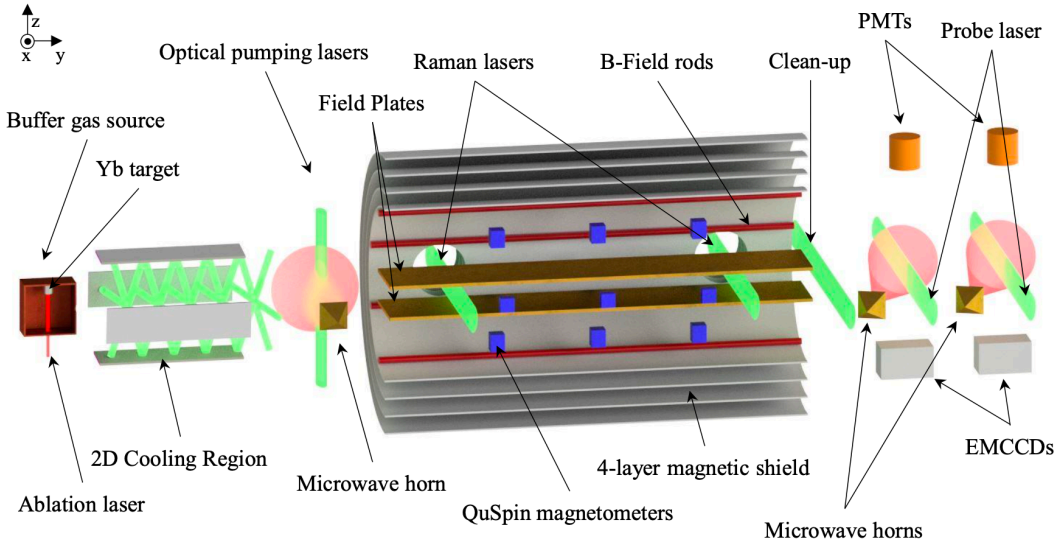


Figure 2.3: Simplified schematic of the ultracold YbF experiment. Figure from [24].

2.3.1 Buffer Gas Source

The buffer gas source produces a slow, bright beam of YbF molecules. A Yb target is mounted inside a copper cell and ablated by a high-power Nd:YAG laser pulse. The material then interacts with a

continuous flow of SF_6 that is injected into the cell to form YbF. A separate flow of He gas cools the YbF through collisions before the molecules leave through a 4 mm aperture. The entire cell itself is coated in charcoal to absorb the excess He. The source produces a flux of $>10^{10}$ molecules per steradian mostly in the $N = 1$ state at a transverse temperature comparable to that of the He gas which is thermalised with the cell at 3.8 K. The resulting velocity distribution has a mean of 175 ms^{-1} and standard deviation of 40 ms^{-1}. The time of flight is defined as the interval between ablation and reaching the detection region. Data is taken over 300 shots at repetition rate of 5 Hz.

2.3.2 Transverse Sisyphus Cooling

The sensitivity of the experiment scales as $1/\sqrt{N_{\text{mol}}}$, so maximising N_{mol} is essential. Here, Sisyphus cooling in a 20-cm-long cooling region reduces the transverse temperature to below 100 µK. The cooling operates in the rotationally closed $X^2\Sigma^+(v = 0, N = 1) \rightarrow A^2\Pi_{1/2}, (v = 0, J = 1/2, e)$ transition via P(1) light in figure 2.1. Since the ground state contains higher angular momentum than the excited state, it includes dark states which do not optically couple to the laser light [25]. For example, for z-polarised light, the stretched Zeeman hyperfine levels in the ground state are dark because a transition could not conserve angular momentum. The magnetically assisted Sisyphus effect (MASE) exploits these dark states. The cooling light forms a standing wave along the direction of cooling and is blue-detuned so that $\Delta = \omega - \omega_0 > 0$ where ω_0 is the resonant frequency. A molecule then experiences an intensity dependant AC stark shift of its potential [26]

$$U(z) \approx \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}I(z) \tag{2.8}$$

where Γ is the spontaneous decay rate of the excited state. The shift is approximated by the two-level result which holds well in a rotationally closed transition. A molecule in a bright state therefore climbs a potential hill that peaks at the intensity maximum while dark states do not experience an energy shift [27]. As bright molecules states approach the intensity maximum, the scattering rate

$$\Gamma_{\text{sc}} \propto I(r) \tag{2.9}$$

is maximised, so this is where it is most likely to be pumped into dark states. The molecule, now dark, continues across the flat potential to the region of a node where a magnetic field remixes it into a bright state via Larmor precession, and the cycle of climbing the potential and dissipating energy begins again [28].

2.3.3 Optical Pumping

Cooling leaves the molecules distributed across the $X^2\Sigma^+(v = 0, N = 1)$ manifold, whereas the eEDM measurement is prepared from $X^2\Sigma^+(v = 0, N = 1, F = 0)$. The population must therefore be transferred so they do not contribute to the background. Microwaves (pink in figure 2.1) first drive the molecules to $X^2\Sigma^+(v = 0, N = 2)$ from where an optical laser drives the ${}^oP_{12}(2)$ transition to $A^2\Pi_{1/2}, (v = 0, J = 1/2, f)$. From there, molecules may decay to $|N = 0, F = 0\rangle$ and $|N = 0, F = 1\rangle$, with some loss to higher vibrational states. Since the population should accumulate only in $F = 0$, an additional $Q(0)$ laser transitions empties the $F = 1$ state. Optical cycling continuous until all molecules have either accumulated in the desired $F = 0$ state or have been lost to $v>0$. This pumping transition is displayed in figure 2.1.

2.3.4 Interaction Region

In the ultracold eEDM experiment, the measurement is taken from the state

$$|x\rangle = \frac{1}{\sqrt{2}}\left(|F = 1, m_F = +1\rangle + |F = 1, m_F = -1\rangle\right). \tag{2.10}$$

Stimulated Raman Adiabatic Passage (STIRAP) is used to prepare and later read out quantum superposition states, which transfers populations between two ground states via an excited state without populating it and hence without spontaneous emission [29, 30]. Here, the initial state is $|X^2\Sigma_{1/2}(N = 0, F = 0)\rangle$, the target is $|x\rangle$, and the intermediate state is $\left|A^2\Pi_{1/2}(J = 1/2, F = 1)\right\rangle$.

With the molecules prepared in $|x\rangle$, they are left to evolve in the interaction region which is the core of the experiment where spin precession interferometry takes place. Both electric and magnetic fields are applied here. The electric field plates are a distance of 1.2 cm apart and a ±6 kV potential is applied with high-voltage modules, generating a uniform electric field of 10 kVcm^{-1} which produces a polarisation factor $\eta = 0.557$. The electric field direction can be reversed. A magnetic field is also applied using two pairs of coils mounted on the inside of the magnetic shielding producing a field of about 20 nT. Under the combiend fields, the molecule gains a phase

$$\begin{aligned} |x(t)\rangle &= \frac{1}{\sqrt{2}}\left(e^{-i\phi}|F=1, m_F=+1\rangle + e^{i\phi}|F=1, m_F=-1\rangle\right) \\ &= \cos(\phi)|x\rangle + \sin(\phi)|y\rangle \end{aligned} \tag{2.11}$$

where we have defined an orthogonal state $|y\rangle = \frac{i}{\sqrt{2}}\left(|F=1, m_F=+1\rangle - |F=1, m_F=-1\rangle\right)$ such that $\langle x|y\rangle = 0$. Now the phase is

$$\phi = \phi_E + \phi_B \tag{2.12}$$

$$\phi_E = -d_e m_F \eta \varepsilon_{\text{eff}} \tau/\hbar \tag{2.13}$$

$$\phi_B = \mu_B g_F m_F m_F B\tau/\hbar \tag{2.14}$$

where τ is the spin precession time. STIRAP is finally used to map $|x\rangle$ back to $|F=0\rangle$ leaving $|y\rangle$ in $|F=1\rangle$. The difference between these two states is the key for measuring the eEDM.

2.4 Detector

2.4.1 Principle of Detection

The eEDM measurement is read out through the populations in $|N=0, F=1\rangle$ in detector A and $|N=0, F=0\rangle$ in detector B which are spatially separated. Microwaves couple each to the $N=1$ manifold from where two counter-propagating lasers driving the P(1) optical cycling transition connects $X^2\Sigma^+(v=0, N=1) \rightarrow A^2\Pi_{1/2}(v=0, J=1/2, e)$ shown in figure 2.1, and the fluorescence photons from spontaneous decays are recorded. The time of flight (TOF) is the time taken for the molecules to travel from ablation to the the detection volume. Integrating the TOF curves gives the number of photons recorded in each detector S_A and S_B. Using equation 2.11, these are

$$\begin{aligned} S_A &= N\epsilon_A \cos^2(\phi) \\ S_B &= N\epsilon_B \sin^2(\phi) \end{aligned} \tag{2.15}$$

where ϵ_i is the detection efficiency of detector i and N is the number of molecules. An asymmetry curve is then constructed

$$\mathcal{A} = \frac{S_A - S_B}{S_A + S_B} \approx \mathcal{C}\cos(\phi) \tag{2.16}$$

where $\mathcal{C}$ is the contrast. Extracting d_e from the asymmetry relies on three binary parameters called switches, each taking a value of +1 or -1. $\tilde{E}$ controls the direction of the E-field, $\tilde{B}$ the direction of the B-field, and $d\tilde{B}$ the direction of a small step in B. The green curve in figure 2.4 is an asymmetry curve with $\tilde{E}=+1$ and the blue curve with $\tilde{E}=-1$. Reversing the B-filed from $\tilde{B}=+1$ to $\tilde{B}=-1$ reverses the phase ϕ_B. The small step $d\tilde{B}$ is under direct experimental control and corresponds to exactly

$$\phi_{\delta B} = \frac{g_F \mu_B B_{\delta B}\tau}{\hbar} = \frac{\pi}{32} \tag{2.17}$$

which allows the asymmetry curve to be sampled at points displaced along it. Switching the electric field with $\tilde{E}$ is sensitive to d_e, shifting the curve by $2\phi_E$. To extract a measurement, a channel is defined as

$$\{X\} = \frac{1}{N}\sum_i \tilde{X}_i \mathcal{A}_i \tag{2.18}$$

which is the weighted average of asymmetry $\mathcal{A}_i$ and switch $\tilde{X}_i$ over the i-th measurement. Channels can also be used for combined switches. For example, it is straightforward to show that

$$\begin{aligned} \{E\cdot B\} &\approx 2\mathcal{C}\phi_E \\ \{\delta B\} &\approx 2\mathcal{C}\phi_{\delta B} \end{aligned} \tag{2.19}$$

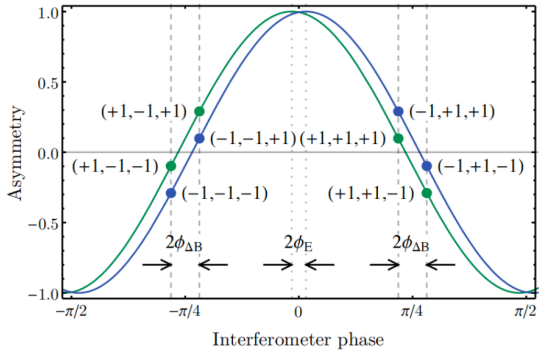


Figure 2.4: Asymmetry curve against phase caused by B-field. The values in brackets are the switches $(\tilde{E}, \tilde{B}, d\tilde{B})$. These 8 switches are combined into a *data block*.

v''	**Probability**
0	0.933 ± 0.003
1	0.066 ± 0.003
2	$(3 \pm 0.5) \times 10^{-3}$
3	$0^{+5}_{-0} \times 10^{-4}$

Table 2.2: Frank-Condon factors determining the branching fractions of the $A^2\Pi_{1/2}(v' = 0) \rightarrow X^2\Sigma^+(v'')$ decay [31].

which is used to determine

$$d_e = \frac{\hbar\phi_E}{\eta\varepsilon_{\text{eff}}\tau} = \frac{\hbar\phi_{\delta B}}{\eta\varepsilon_{\text{eff}}\tau}\frac{2\mathcal{C}\phi_E}{2\mathcal{C}\phi_{\delta B}} = \frac{\hbar\phi_{\delta B}}{\eta\varepsilon_{\text{eff}}\tau}\frac{\{E \cdot B\}}{\{\delta B\}}. \tag{2.20}$$

Switches and channels are used to automate data acquisition.

2.4.2 Detection Apparatus

Photomultiplier tubes (PMTs) provide temporal information, while a Andor iXon 888 CCD camera adds spatial resolution. The PMT and CCD camera are on opposite sides of the beamline. Ideally, molecules cycle in the P(1) transition until they get lost to higher vibrational states. The branching fraction to higher vibrational levels is given by the Frank-Condon factors listed in table 2.2. With a branching fraction back to $v = 0$ of 0.933, cycling to completion would give an average of 14.9 photons per molecule. In practice, around 13.4 are scattered in transit.

The microwaves are generated by the *Windfreak SynthHD* and enter via a microwave horn angled at 45°. In detector A, the $|N = 0, F = 1\rangle \rightarrow |N = 1, F = 2\rangle$ transition at 14.458 GHz was originally used, since it has the highest transition strength, and being a type-I transition, supports no dark states. This was later changed to $|N = 0, F = 1\rangle \rightarrow |N = 1, F = 1^+\rangle$ at 14.489 GHz, out of concern that the energetically similar transition of $|N = 0, F = 0\rangle$ might also be driven. Because the Rabi-frequencies of the hyperfine levels all differ, driving the $F = 2$ state leaves roughly 50% of the molecules in $F = 2$ and 50% in $N = 0$. This drops to about 33% when driving $F = 1^+$ due to an additional dark state in the Type-II transition. However, experimental data showed the resulting loss of signal to be minimal. In detector B, the microwaves drive $|N = 0, F = 0\rangle \rightarrow |N = 1, F = 1^-\rangle$ at 14.467 GHz. Since the TOF profiles between detectors A and B overlap subsantially, the two detection chambers need to be shielded from one another using a thin metal mesh with 4.9 mm diameter holes with a 2 cm central hole for the beamline. The chamber interior is lined with microwave absorbing *Laird Eccosorb HR* foam to strongly damp scatter and reflection which ideally leaves the microwaves coherent and uniform.

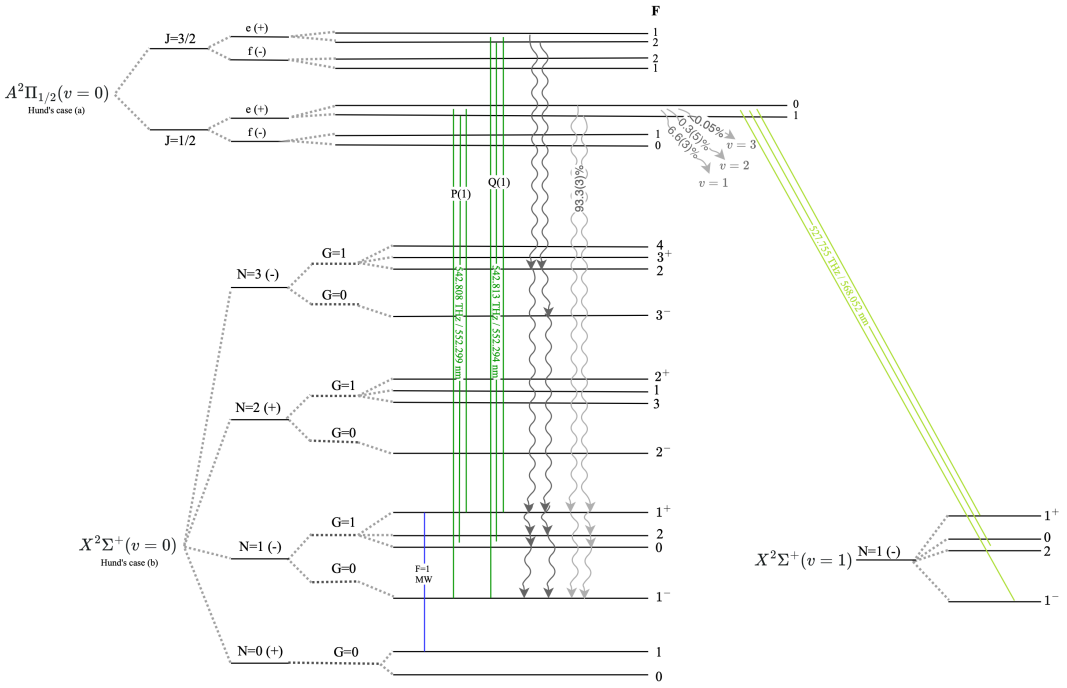


Figure 2.5: Detection P(1) laser with clean-up laser Q(1) and re-pump laser V1. The microwave transition (blue) in detector A is also shown.

2.4.3 Signal Model and Detection Imperfections

There are two three types of noise in the detector. Denoting P_1 as the particles in state $F = 1$ in detector A and P_0 for state $F = 0$ in detector B, the measured signals are [24]

$$S_A = (P_1 + x_A P_0 + P_{BG})\varepsilon_A \tag{2.21}$$

$$S_B = ((1 - x_A)P_0 + x_B P_1)\varepsilon_B. \tag{2.22}$$

x_A is the fraction of P_0 detected in region A due to microwave leakage from cavity B into region A causing molecules to be cycled in the wrong detector and producing a false signal. P_{BG} is the background scattering, i.e. molecules which have been left in $|N = 1\rangle$ or others from upstream components. Finally, x_B is the probability of detecting P_1 in detector B due to incomplete cycling. Some (particularly faster molecules) do not scatter enough photons to be lost to $v > 0$ and instead remain in $X^2\Sigma^+(N = 1, v = 0)$. Upon reaching detector B, they scatter resonant photons via the P(1) laser, contributing to a false signal and reducing contrast. For this experiment, $x_B \sim 0.05$.

2.4.4 Statistical Sensitivity

These three types of noise contribute to decease the eEDM sensitivity. The statistical uncertainty in a phase measurement is

$$\sigma_\phi = \frac{1}{2\sqrt{N\varepsilon}}, \tag{2.23}$$

referred to as the "photon shot noise limit" which reduces to the fundamental quantum projection limit for $\varepsilon = 1$. When employing the optical cycling transition P(1) in detector A with a branching fraction b_l to dark states (here $v > 0$), the uncertainty gains an additional excess noise factor F such that [32]

$$\sigma_\phi = \frac{1}{2\sqrt{N}}\sqrt{F}. \tag{2.24}$$

This factor is fundamental and can be calculated as

$$F = 1 + \frac{1}{\bar{n}}\left(\frac{1}{\epsilon} - 1\right) + \frac{\sigma_n^2}{\bar{n}^2} \tag{2.25}$$

where $\bar{n}$ is the average number of photons scattered per molecule and σ_n is its standard deviation. ϵ is the probability of detecting a photon, including geometric factors and detection efficiency. From this formula it can be shown that if the signal in detector A is maximised by cycling to completion which corresponds to the limit $b_l rT \to \infty$ where r is the scattering rate per optical cycle and T is the average interaction time, then the uncertainty increases by a factor of $\sqrt{F} = \sqrt{2}$ relative to the ideal quantum shot noise. Hence, cycling to completion minimises x_B by fully depleting the $|N = 1\rangle$ population, but incurs excess noise through F. However, $F \approx 1$ can be recovered by operating at a finite cycling time rT that simultaneously satisfies $rT\varepsilon \gg 1$ so that many photons per molecule are detected and $rTb_l \ll 1$ so that the probability of decaying to a dark state is small. These two conditions are compatible if b_l/ε is small which motivates reducing b_l. Adding a V1 laser which is demonstrated in figure 2.5 in light-green to re-pump molecules lost to $v = 1$ back to $|v = 0, N = 1\rangle$ achieves this since decays to $v \geq 2$ are rare by the Franck-Condon factors [31] in table 2.2 and hence b_l is minimised. However, molecles which are re-pumped scatter for much longer and some residual population will remain in the $|v = 0, N = 1\rangle$ state even after the laser beam. This will significantly increase x_B. Therefore, another clean-up laser Q(1) is introduced in figure 2.5 which pumps molecules from $N = 1$ to $|A^2\Pi_{1/2} J = 3/2, e\rangle$ from which they may decay to $|N = 1,\ N = 3\rangle$, or higher vibrational states, hence accumulating in $N = 3$ over time. This would remove the x_B noise while allowing the V1 laser to reduce F and increase the sensitivity.

Equation 2.25 requires an advanced simulation to estimate the correct mean and standard deviation of the number of photons scattered per molecule. For this reason, a computer simulation of the detector scheme was developed including both the P(1) and V1 lasers to estimate how much F could be improved with this new detection scheme. Additionally, the simulation can guide investigations in how to maximise the number of photons scattered $\bar{n}$. This requires an exhaustive understanding of the detection lasers and their sidebands which are described below.

2.4.5 Detector Laser Details

To drive the transitions given above, the P(1) laser operates at 552 nm, and the V1 laser operates at 568 nm. The intensity shape is a gaussian with standard deviation of 2.45 mm (9.8 mm full width at $1/e^2$).

Electro-Optic Modulators (EOMs) and Acousto-Optic Modulators (AOMs) generate the sidebands addressing all hyperfine levels. The EOM uses a birefringent $LiNbO_3$ crystal doped with 5% MgO. Applying a small external electric field E_z changes the refractive index [30, 33]

$$\Delta n \approx -\frac{1}{2} n_0^3 r_{33} E_z. \tag{2.26}$$

where n_0 is the refractive index with no electric field and r_{33} is the electro-optic tensor for a refractive index change along z when the electric field is also applied along $\hat{z}$. A light field travelling in $\hat{z}$ will experience a phase shift

$$\phi = \frac{\pi n_o^3 r_{33} V}{\lambda} \frac{l}{d} \tag{2.27}$$

where l is the length of the crystal, d is the separation between plates across which a voltage V is applied. A sinusoidal drive $V(t) = V_0 \sin(\omega_0 t)$ will then induce a time-dependent phase

$$\phi(t) = \frac{\pi n_o^3 r V_0}{\lambda} \frac{l}{d} \sin(\omega_0 t) = \phi_0 \sin(\omega_0 t) \tag{2.28}$$

so that the E-field of the light becomes

$$E(t) = E_0 e^{-i\omega t + \phi_0 \sin(\omega_0 t)}. \tag{2.29}$$

This is readily rewritten using the Jacobi-Anger identity [23]

$$E(t) = E_0 e^{i\omega t} \left(J_0(\phi_0) + \sum_k J_k(\phi_0) e^{ik\omega_0 t} + i \sum_k (-1)^k J_k(\phi_0) e^{-ik\phi_0 t} \right) \tag{2.30}$$

where J_k are Bessel functions of k-th order. The terms can be interpreted as sidebands in the frequency spectrum, with frequencies $\omega \pm k\omega_0$ and amplitudes $E_0 J_k(\phi_0)$. The orders $k = 0, \pm 1$ are the most prominent with the amplitude of higher order sidebands quickly dropping off. Different waveforms $V(t)$ can

be used to obtain different sideband spectra.

EOMS can also be used to modulate the polarisation. Rotating the polarisation by 90° requires the crystal's z-axis to be at 45° to the input polarisation. Consider, for example, a light beam travelling along x with equal polarisation components in y and z. These two components will experience a different refractive index

$$\Delta n_y = \frac{1}{2} n_o^3 r_{13} E_z \tag{2.31}$$

$$\Delta n_z = \frac{1}{2} n_e^3 r_{33} E_z \tag{2.32}$$

where n_o is the ordinary refractive index and n_e is the extraordinary refractive index, constants for a given material. The components hence accumulate a relative phase

$$\Delta\phi = \frac{\pi l V}{\lambda d}(n_e^3 r_{33} - n_o^3 r_{13}). \tag{2.33}$$

Adjusting V allows $\Delta\phi = \pi$ for example. In practice, the EOM is driven with a linearly increasing retardance ramp $V(t)$, so that polarisation cycles repeatedly through *linear in* $y \rightarrow \sigma^+ \rightarrow$ *linear in* $z \rightarrow \sigma^-$. The EOM is temperature stabilised to minimise random drifts in ϕ. The P(1) and V1 laser transitions are polarisation modulated at 1 MHz.

Alongside the EOM, AOMs are used to generate sidebands. Where EOMs alter the refractive index by applying an electric field, AOMs use the acousto-optic effect where change in mechanical strain a causes a change in the optical properties of the crystal. A plane acoustic wave with wavelength Λ is applied across the AOM propagating along z. These pressure fluctuations then cause a change in the refractive index [34]

$$n(z,t) = n_0 + \Delta n \cos(\Omega t - Kz) \tag{2.34}$$

where $K = 2\pi/\Lambda$ is the wave-vector in the material, Ω is the angular frequency, and n_0 is the undisturbed refractive index. Light passing through the crystal encounters a diffracting grating moving at the speed of sound and must satisfy the Bragg condition

$$\Lambda \sin(\theta_m) = m\lambda \tag{2.35}$$

where λ is the wavelength of the incoming light and $m \in \mathbb{N}$. The ultracold experiment operates operating at the Klein-Cook parameter regime $Q \gg 1$, reached with high acoustic frequencies and longer crystals [34]. Here, the first $m = \pm 1$ dominate strongly and higher orders vanish. As the incoming light is incident close to the Bragg angle and scatters of phonons, it experiences a doppler shift

$$\omega_f = \omega_i \pm \Omega. \tag{2.36}$$

The AOM hence imparts a single, coherent frequency shift of the entire sideband exact to the drive's own stability and precision.

The EOMs and AOMs as described are used to cover the hyperfine levels of YbF. The $X^2\Sigma^+(v = 0, N = 1)$ manifold has 4 F states: $F = 1^-, 0, 2, 1^+$ in ascending order of energy with relative detunings 0, 155.7, 161.2, 192.1 MHz respectively. The fundamental P(1) laser light is split into three distinct beams. The fundamental beam drives the $X^2\Sigma^+(N = 1, F = 1^-)$ transition and is unshifted. Two AOMs shift the remaining two beams using the $m = -1$ order: one by 192 MHz to address the $F = 1^+$ state, and the other by 158 MHz to address both $F = 0$ and 2 simultaneously, since these lay close together. This is illustrated in figure 2.6.

As part of this project, an additional V1 laser was added to re-pump molecules from higher vibrational levels back to $v = 0$. The $X^2\Sigma^+(v = 1, N = 1)$ manifold likewise contains 4 F states: $F = 1^-, 0, 2, 1^+$ with relative energies 0, 151.8, 176.5, 205.8 MHz. The V1 laser fundamentally drives the $X^2\Sigma^+(N = 1, F = 2)$ transition. This is split into two beams, one of which is shifted by an AOM at $m = 1$ order to $F = 1^-$ with 170 MHz. The $F = 2$ beam obtains sidebands at ± 27 MHz to address $F = 1^-$ and $F = 0$ as seen in figure 2.6.

Diffraction efficiency, mirror losses, the efficiency and configuration of the EOMs and AOMs, and other loss mechanisms mean that the power in each sideband may differ. This was measured: While the P(1)

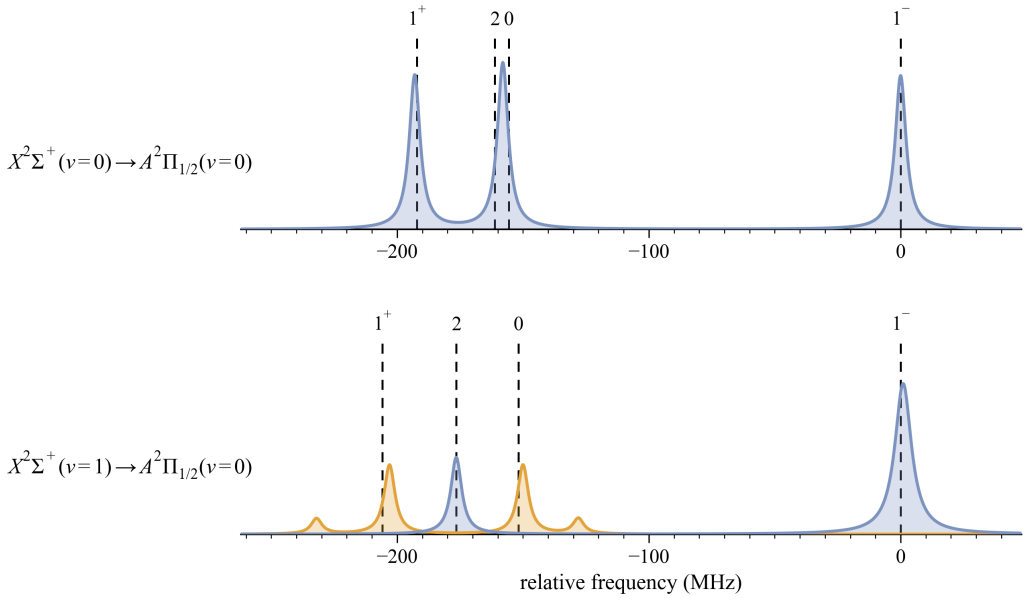


Figure 2.6: Hyperfine level structure with laser frequency spectrum and sideband coverage. Blue is AOM shifted and yellow is EOM shifted. The peak height shows approximate relative power. The dotted lines are the true sideband energies. Figure adapted and updated from [33].

laser sidebands each carry 1/3 of the total power, the $F = 1^-$ sideband carries 43% power while the other three sidebands have 19% power each. The approximate relative sideband energies are displayed in figure 2.6. The linewidth of each P(1) sideband is 754 kHz, and for each V1 sideband 541 kHz.

The laser setup as used in the detectors A and B was implemented in the simulation code with the fidelity described above.

3

Monte-Carlo Wavefunction Simulation

3.1 Choice of Framework

A number of computational methods have been developed to simulate quantum optical systems [35, 36]. Rate equation models treat the energy levels as reservoirs connected by stimulated emission, absorption and spontaneous emission, with the transfer described by rate coefficients that are often determined empirically. Their simplicity, efficiency and applicability to both open and closed systems make them a natural first choice. However, due to their semi-classical approach, they treat all population transfer as incoherent and can therefore not represent coherent dark states. In the ultracold experiment, some sidebands drive a type-II transition ($F \geq F'$), so molecules are pumped into coherent superpositions that cease to scatter. For the detection region, rate equations therefore significantly overestimate the number of photons scattered and cannot be used with accuracy. They also offer no route to higher-order effects such as polarisation modulation or temperature-dependent behaviour. The Monte-Carlo wavefunction (MCWF) method, by contrast, is a full quantum mechanical evolution of a wavefunction with random quantum jumps, each corresponding to the emission of a single photon. This is faster than optical bloch equations, which propagate the entire density matrix ρ and therefore carry $\mathcal{O}(N^2)$ elements rather than $\mathcal{O}(N)$, and it gives access to quantities that are natural to single quantum systems, such as full photon statistics and emission timing. Unlike steady-state approaches, it also permits time-dependent control over parameters such as phase jitter and polarisation modulation. Together these features allow a full quantum description of the detector setup.

3.2 Mathematical Description

3.2.1 Rotating Wave Approximation (RWA)

At the heart of most quantum optical simulations, MCWF included, lies the rotating wave approximation (RWA). First, a simple two-state system is presented and then it is extended to include hyperfine splitting and multiple-laser driven transitions in full generality.

The atomic hamiltonian in its eigenbasis can be written

$$\hat{H}_A = E_g|g\rangle\langle g| + E_e|e\rangle\langle e| \tag{3.1}$$
$$= E_g\hat{1} + \omega_0|e\rangle\langle e| \tag{3.2}$$

with ground state energy E_g and excited state energy E_e and setting $\hbar = 1$. We have also defined $\omega_0 = E_e - E_g$ as the energy difference between the excited and ground state. The interaction occurs through the electric dipole operator

$$\hat{d} = d\left(|e\rangle\langle g| + |g\rangle\langle e|\right) \tag{3.3}$$

which connects the ground and excited state allowing for excitations and decays. Considering only classical fields, the interaction term of the hamiltonian is

$$\hat{H}_{\text{int}} = -\hat{d}\cdot\mathbf{E}(t), \tag{3.4}$$

we can write the full hamiltonian as

$$\hat{H} = \hat{H}_A + \hat{H}_{\text{int}} \tag{3.5}$$
$$= \omega_0|e\rangle\langle e| + \Omega\cos(\omega_L t)\left(|e\rangle\langle g| + |g\rangle\langle e|\right) \tag{3.6}$$
$$= \omega_0|e\rangle\langle e| + \frac{\Omega}{2}\left(e^{i\omega_L t} + e^{-i\omega_L t}\right)\left(\sigma^+ + \sigma^-\right) \tag{3.7}$$

where ω_L is the laser frequency and assuming $\mathbf{d} \parallel \mathbf{E}$ which allows us to define the rabi frequency $\Omega = -dE_0$ and the constant energy offset $E_g\hat{1}$ has been dropped. The matrices are $\sigma^+ = |e\rangle\langle g|$ and $\sigma^- = |g\rangle\langle e|$. We seek a frame of reference in which the fast optical phases are removed. Since the laser drive enters as a phase $e^{\pm i\omega_L t}$ on $\sigma^\pm$, we take a diagonal unitary operator

$$U = e^{i\theta t|e\rangle\langle e|} \tag{3.8}$$

which is justified by considering its conjugation on σ^+ and σ^-:

$$U\sigma^+U^\dagger = e^{i\theta t}\sigma^+ \qquad U\sigma^-U^\dagger = e^{-i\theta t}\sigma^- \tag{3.9}$$

only multiplies $\sigma^\pm$ by a phase. θ is a free parameter and any choice is equally valid. Transforming with $\tilde{H} = U\hat{H}U^\dagger + i\dot{U}U^\dagger$ then gives

$$\tilde{H} = (\omega_0 - \theta)|e\rangle\langle e| + \frac{\Omega}{2}(e^{-i(\omega_L-\theta)t}\sigma^+ + e^{+i(\omega_L-\theta)t}\sigma^-) + \frac{\Omega}{2}\left(e^{i(\omega_L+\theta)t}\sigma^+ + e^{-i(\omega_L+\theta)t}\sigma^-\right) \tag{3.10}$$

which is exact. The RWA lies in recognising that in any meaningful time-period, the dynamics of very fast oscillating terms average to 0 since [37]

$$\int e^{\pm in\theta}d\theta \approx 0 \tag{3.11}$$

for large n. For any θ within a few GHz of ω_L, then $\omega_L + \theta \gg \omega_L - \theta$ means the third term in equation 3.10 is discarded. There are two choices of interest: Setting $\theta = \omega_L$, the hamiltonian simplifies to

$$\tilde{H} \approx -\delta|e\rangle\langle e| + \frac{\Omega}{2}(\sigma^+ + \sigma^-) \tag{3.12}$$

in which the detuning $\delta = \omega_L - \omega_0$ sits on the diagonal and the coupling is static. Setting instead $\theta = \omega_0$, i.e. rotating at the molecular rather than laser frequency, gives

$$\tilde{H} \approx \frac{\Omega}{2}(e^{-i\delta t}\sigma^+ + e^{i\delta t}\sigma^-) \tag{3.13}$$

in which the excited state carries no detuning and all the detuning appears as a phase on the coupling. This convention is the one that is adapted from here on as it is the most convenient for simulating molecules. It should be noted that the two forms are equivalent and connected by a diagonal unitary matrix. Having worked through the simple 2-state case, the RWA is now extended to include arbitrarily many states.

YbF is not a two-state molecule. The simulation contains the four vibrational ground manifolds $X^2\Sigma^+(v = 0, 1, 2, 3;\ N = 1)$, the excited manifold $A^2\Pi_{1/2}(v' = 0,\ J' = 1/2)$, and optionally the $X(N = 0)$ manifold addressed by microwaves. Each carries its own hyperfine structure. Label a manifold m and define the set of projectors

$$\hat{P}_m = \sum_{F,m_F \in m} |mFm_F\rangle\langle mFm_F|. \tag{3.14}$$

This extends the atomic hamiltonian to a sum over the manifold origin and the internal structure relative to that origin [38]

$$H_A = \sum_m \left(E_m \hat{P}_m + \hat{h}_m\right) \tag{3.15}$$

where E_m is the lab frame energy of the manifold centre which includes electronic, vibrational, and rotational shifts. $\hat{h}_m$ is the internal shift, i.e. spin-rotation, hyperfine, and Zeeman splitting. It is defined

$$\hat{h}_m = \sum_{F,m_F \in m} \omega^{(m)}_{F,m_F} |mFm_F\rangle\langle mFm_F| \tag{3.16}$$

where $\omega^{(m)}_{F,m_F}$ is the energy of relative to E_m. For example, for a ground manifold X and excited manifold A whose (arbitrarily chosen) centres are separated by ω_0, this reads

$$H_A = \sum_{F,m} \omega^X_{F,m_F} |XFm_F\rangle\langle XFm_F| + \sum_{F',m'_F} (\omega_0 + \omega^A_{F',m'_F})|AF'm'_F\rangle\langle AF'm'_F| \tag{3.17}$$

where ω^X_{F,m_F} and ω^A_{F',m'_F} are the frequency offsets. For the RWA to be applicable, this requires $\omega^X_{F,m_F}, \omega^A_{F',m'_F} \ll \omega_L$, To define the electric dipole operator, we may no longer assume $\mathbf{d} \parallel \mathbf{E}$. Instead, we split it into two parts [18]

$$\hat{\mathbf{d}} = \hat{1}\hat{\mathbf{d}}\hat{1} = \left(\sum_m \hat{P}_m\right) \hat{\mathbf{d}} \left(\sum_n \hat{P}_n\right) = \hat{\mathbf{d}}^+ + \hat{\mathbf{d}}^- \tag{3.18}$$

with $\hat{\mathbf{d}}^- = (\hat{\mathbf{d}}^+)^\dagger$. The diagonal pieces $\hat{P}_m \hat{\boldsymbol{d}} \hat{P}_m$ average to 0 because the dipole operator is parity odd and each manifold has definite parity. From now on, we will be working in the spherical helicity basis [30] where

$$\hat{e}_{\pm 1} = \mp \frac{1}{\sqrt{2}}(\hat{x} \pm i\hat{y}), \qquad e_0 = \hat{z} \tag{3.19}$$

with inner product

$$\boldsymbol{A} \cdot \boldsymbol{B} = \sum_q (-1)^q A_q B_{-q}. \tag{3.20}$$

Taking the polarisation unit vector of the photon to be $\boldsymbol{\epsilon}$, then the inner product of the dipole operator with the polarisation is

$$\hat{\mathbf{d}} \cdot \boldsymbol{\epsilon} = \sum_q (-1)^q \epsilon_q \hat{d}_{-q}. \tag{3.21}$$

The raising part of the dipole operator defines the coupling operators $\hat{\Sigma}_q$. By definition [39, 40]

$$\hat{\boldsymbol{d}}^+ = \sum_q \hat{e}_q \left(\hat{P}_A \hat{d}_q \hat{P}_X\right) \equiv d_0 \sum_q \hat{e}_q \hat{\Sigma}_q \tag{3.22}$$

$$\hat{\boldsymbol{d}}^- = \sum_q \hat{e}_q \left(\hat{P}_X \hat{d}_q \hat{P}_A\right) \equiv d_0 \sum_q (-1)^q \hat{e}_q \hat{\Sigma}^\dagger_{-q} \tag{3.23}$$

where d_0 is a normalisation constant and $\hat{\Sigma}_q$ is the hyperfine equivalent of σ^+. Explicitly, $\hat{\Sigma}_q$ can be found using the Wigner-Eckart theorem

$$\hat{\Sigma}_q = \sum_{F,m_F} \sum_{F',m'_F} \frac{1}{d_0} \langle F'||d||F\rangle \frac{\langle F, m_F; 1, q|F', m'_F\rangle}{\sqrt{2F'+1}} |AF'm'_F\rangle\langle XFm_F| \tag{3.24}$$

and it connects all hyperfine levels with the appropriate coupling strength. The normalisation constant d_0 is fixed by requiring that the total spontaneous decay rate is Γ, so

$$\sum_q \hat{\Sigma}_q^\dagger \hat{\Sigma}_q = \hat{P}_A. \tag{3.25}$$

Each transition is driven by multiple lasers and EOM and AOM shifted sidebands. For L frequency laser components, the field with polarisation vector $\boldsymbol{\epsilon}$ is [38]

$$\boldsymbol{E}(t) = \sum_{l=1}^{L} \frac{E_l(\boldsymbol{r})}{2} \left(\boldsymbol{\epsilon}^{(l)} e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega_l t)} + \boldsymbol{\epsilon}^{(l)*} e^{-i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega_l t)} \right) \tag{3.26}$$

where $E(\boldsymbol{r})$ is the real envelope, gaussian in the ultracold experiment. The interactive hamiltonian is [39]

$$H_{\text{int}} = -\hat{\mathbf{d}} \cdot \boldsymbol{E}(t) = -\sum_l \frac{E_l(\boldsymbol{r})}{2} \left((\boldsymbol{d}^+ + \boldsymbol{d}^-) \cdot \boldsymbol{\epsilon}^{(l)} e^{i(\boldsymbol{k}_l\cdot\boldsymbol{r}-\omega_l t)} + (\boldsymbol{d}^+ + \boldsymbol{d}^-) \cdot \boldsymbol{\epsilon}^{(l)*} e^{-i(\boldsymbol{k}_l\cdot\boldsymbol{r}-\omega_l t)} \right) \tag{3.27}$$

Given the definition of $\hat{\Sigma}_q$ in equation 3.22 and setting $\Omega_l = -d_0 E_l(\boldsymbol{r})$, the interaction hamiltonian becomes

$$\begin{aligned} H_{\text{int}} = &\sum_{l,q} \frac{\Omega_l}{2} \left(e^{+i(\boldsymbol{k}_l\cdot\boldsymbol{r}-\omega_l t)} (-1)^q \epsilon_{-q}^{(l)} \hat{\Sigma}_q + e^{-i(\boldsymbol{k}_l\cdot\boldsymbol{r}-\omega_l t)} (-1)^q \epsilon_{-q}^{(l)*} \hat{\Sigma}_q^\dagger \right) \\ &+ \sum_{l,q} \frac{\Omega_l}{2} \left(e^{+i(\boldsymbol{k}_l\cdot\boldsymbol{r}-\omega_l t)} \epsilon_q^{(l)} \hat{\Sigma}_q^\dagger + e^{-i(\boldsymbol{k}_l\cdot\boldsymbol{r}-\omega_l t)} \epsilon_q^{(l)*} \hat{\Sigma}_q \right) \end{aligned} \tag{3.28}$$

Now we introduce the multi-state equivalent unitary operator [39]

$$U = \exp\left(it \sum_m \theta_m \hat{P}_m \right) = \sum_m e^{it\theta_m} \hat{P}_m \tag{3.29}$$

for which we make two observations:

1. It subtracts every manifold origin $i\dot{U}U^\dagger = -\sum_m \theta_m \hat{P}_m$
2. Using the property $\hat{P}_m \hat{P}_n = \delta_{mn} \hat{P}_m$, every operator $\mathcal{O}$ connecting manifold m to manifold n acquires a phase only $U(\hat{P}_n \mathcal{O} \hat{P}_m)U^\dagger = e^{i(\theta_n - \theta_m)t} \hat{P}_n \mathcal{O} \hat{P}_m$ identically to the simple two-state case of the RWA above.

For the atomic hamiltonian, propery 2. means that $U\hat{h}_m U^\dagger = e^{i(\theta_m-\theta_m)t}\hat{h} = \hat{h}$ since $\hat{h}$ connects manifold m to itself. Hence, if we choose to rotate at each manifolds natural frequency $\theta_m = E_m$, then only the internal structure of the atomic hamiltonian remains

$$\tilde{H}_A = \sum_m \hat{h}_m. \tag{3.30}$$

This is the static part of the hamiltonian in the simulation. It only contains its hyperfine and Zeeman structure. For the interaction part of the hamiltonian, every term carrying $\hat{\Sigma}_q$ will acquire a phase associated with the two manifolds $e^{i(E_A - E_{X_v})t}$. It's total phase is $e^{i(E_A - E_{X_v} - \omega_l)t} = e^{-i\delta_l t}$ with detuning

$$\delta_l = \omega_l - (E_A - E_{X_v}). \tag{3.31}$$

The second terms in equation 3.28 get a phase $e^{i(E_A - E_{X_v} + \omega_l)t} \approx e^{2i\omega_l t}$ and is discarded by the RWA. Finally, we recover [41]

$$\tilde{H} \approx \sum_m \hat{h}_m + \sum_{l,q} (-1)^q \frac{\Omega_l}{2} \left(e^{+i\boldsymbol{k}_l\cdot\boldsymbol{r}} e^{-i\delta_l t} \epsilon_{-q}^{(l)} \hat{\Sigma}_q + e^{-i\boldsymbol{k}_l\cdot\boldsymbol{r}} e^{+i\delta_l t} \epsilon_{-q}^{(l)*} \hat{\Sigma}_q^\dagger \right) \tag{3.32}$$

which is the RWA hamiltonian as used in the simulation. The fast and vibrational scales have been removed from the diagonal by the frame of reference and from the coupling by the detuning. δ_l is the laser detuning relative to the difference of two manifold origins. In practice, it is therefore calculated as

a difference of two energies from $\hat{h}_m$ and E_m cancels.

The envelope and spatial phase are evaluated along the molecular motion $\boldsymbol{r}(t) = \boldsymbol{r}_0 + \boldsymbol{v}t$. This factorises as

$$e^{i\boldsymbol{k}_l\cdot\boldsymbol{r}_0}e^{i(\delta_l+\boldsymbol{k}_l\cdot\boldsymbol{v})t} \tag{3.33}$$

which is a constant phase offset together with the Doppler shift. For beams propagating along $\pm\hat{x}$, only the v_x component contributes. The constant phase $e^{i\boldsymbol{k}_l\cdot\boldsymbol{r}_0}$ carries the standing-wave structure as a beam and its (perfectly) retro-reflection differ only by the sign of $\boldsymbol{k}_l$ and hence the interference pattern is sampled by randomly drawing $\boldsymbol{r}_0$ across one wavelength. For simulation efficiency purposes, it is often convenient to define a block

$$\hat{M}_l = \frac{\Omega_l}{2}\, e^{-i\mathbf{k}_l\cdot\mathbf{r}_0}\sum_q (-1)^q \epsilon^{(l)}_{-q}\,\hat{\Sigma}^\dagger_q, \qquad \hat{M}(t) = \sum_l \hat{M}_l\, e^{i(\delta_l-\mathbf{k}_l\cdot\mathbf{v})t}, \tag{3.34}$$

where M_l is a constant matrix. It enters the coupling together with its hermitian conjugate and it can be calculated once for each molecule only applying the position-dependent phase making this very fast to process.

3.2.2 Monte-Carlo Procedure

In canonical quantum mechanics, a system's time evolution is strictly unitary which means that coherence is never destroyed. However, this requires that all degrees of freedom are fully known. This is not the case in open quantum systems where a system interacts with its environment, for example by spontaneously emitting photons. This loss of information and energy to the surroundings results in decoherence, i.e. a phase decay. The Lindblad equation is a master equation describing open quantum systems by introducing a jump or loss operator C which describes the dissipative dynamics. It is a markovian equation as it is a stochastic sequence of events where each state only depends on the previous state. It is given by [42]

$$\dot{\rho} = -\frac{i}{\hbar}[H,\rho] + \sum_i \left(C_i\rho C_i^\dagger - \frac{1}{2}\left\{C_i^\dagger C_i, \rho\right\}\right) \tag{3.35}$$

where ρ is the density matrix, C_i is the Lindblad or loss operator. Removing the second term would recover the von-Neumann equation which describes a unitary (closed) quantum system. H is the system hamiltonian and all environmental interactions are handled by the second part of equation 3.35. Spontaneous emission is the only relaxation channel coupling the excited manifold to the ground manifold, and proceeds via electric dipole radiation, so the collapse operators must be built from the lowering part of the dipole operator $C_q \propto \hat{P}_X d_q \hat{P}_A = d_0\hat{\Sigma}^\dagger_q$ [43]. For spontaneous emission, the loss operator can be taken to be [44]

$$C_q = \sqrt{\Gamma}\hat{\Sigma}^\dagger_q. \tag{3.36}$$

with decay rate Γ which is assumed to be the ssame for all excited states. The Born-Oppenheimer approximation lets us separate the vibrational and angular part $|X, v, F, m_F\rangle \approx |X, c\rangle \otimes |F, m_F\rangle$, so the matrix element factorises

$$\langle X, v, F, m_F|\hat{\Sigma}^\dagger_q|A, 0, F', m'_F\rangle = \langle X, v, |A, 0\rangle\langle F, m_F|\hat{\Sigma}^\dagger_q|F', m'_F\rangle. \tag{3.37}$$

Defining the dipole operator as built from the independent angular factors with ground state vibrational level v,

$$\hat{P}_{X_v}\hat{d}^\dagger_q\hat{P}_A = d_0\sum_v \langle X, v|A, 0\rangle\hat{\Sigma}^{(v)\dagger}_q, \tag{3.38}$$

then writing $f_v = |\langle X, v|A, 0\rangle|^2$ as the Frank-Condon factor and ignoring the overall phase finally gives

$$C_{v,q} = \sqrt{\Gamma f_v}\hat{\Sigma}^{(v)\dagger}_q. \tag{3.39}$$

Since $\sum_v f_v = 1$ by definition, the completeness relation is $\sum_{q,v} C^\dagger_{v,q}C_{v,q} = \Gamma\hat{P}_e$. Throughout the simulation, we set $\Gamma = 1$. It should be noted that C is not hermitian and thus not an observable. Its behaviour depends on the chosen basis and hence correct behaviour emerges only when averaging over many trajectories.

The MCWF method [42] will be described here and its equivalence to the Lindblad equation is proven in 3.2.3. The loss operator is used to define the non-hermitian effective hamiltonian

$$H_{\text{eff}} = H - \frac{i}{2}\sum_c C_c^\dagger C_c \tag{3.40}$$

and the normalised wavefunction $|\psi(t)\rangle$ at time t is evolved by this effective hamiltonian instead of the usual Schrödinger equation. For a small timestep dt,

$$|\tilde{\psi}(t+dt)\rangle \approx (1 - iH_{\text{eff}}dt)|\psi(t)\rangle. \tag{3.41}$$

Since H_{eff} is non-hermitian, it is not norm preserving. Within one time-step, the norm decreases as

$$\langle\tilde{\psi}(t+dt)|\tilde{\psi}(t+dt)\rangle = \langle\psi(t)|(1 + iH_{\text{eff}}^\dagger dt)(1 - iH_{\text{eff}}^\dagger dt)|\psi(t)\rangle \tag{3.42}$$

$$= 1 - dt\sum_c \langle\psi(t)|C_c^\dagger C_c|\psi\rangle + \mathcal{O}(dt^2) \tag{3.43}$$

to first order. For a single channel c, define

$$\delta p_c = dt\langle\psi(t)|C_c^\dagger C_c|\psi\rangle \geq 0 \tag{3.44}$$

and in total the norm decreases by $\delta p = \sum_c \delta p_c$. For the MCWF to be valid, we require that $dt \ll \delta p$ while $\delta p \ll 1$ itself for any one step. This parameter can be thought of as a jump probability $\Pi_c = \delta p_c/\delta p$. The monte-carlo approach then draws a random number χ uniformly and a quantum jump is said to happen if $\chi < \delta p$, and no jump occurs if $\chi > \delta p$. If no jump occurs, the wavefunction is simply

$$|\psi(t+dt)\rangle = \frac{|\tilde{\psi}(t+dt)\rangle}{\sqrt{\langle\tilde{\psi}|\tilde{\psi}\rangle}} = \frac{|\tilde{\psi}(t+dt)\rangle}{\sqrt{1-\delta p}} \tag{3.45}$$

and we proceed to evolve it with the effective hamiltonian. Since δp is small, it takes on average many steps until $\chi < \delta p$ and a jump occurs. Once it does, we then use the loss operator to project the wavefunction into its new state

$$|\psi^{(c)}(t+dt)\rangle = \frac{C_c|\psi(t)\rangle}{\sqrt{\langle\psi|C_c^\dagger C_c|\psi\rangle}} = \frac{C_c|\psi(t)\rangle}{\sqrt{\delta p_c/dt}} \tag{3.46}$$

which occurs exactly with probability Π_c.

3.2.3 MCWF-Lindblad Equivalence

The above procedure is a stochastic method which must reproduce the Lindblad equation in the limit of large number of molecules. To see the equivalence, take the standard form of the density operator at some later time with the two possibilities of jump and no-jump

$$\rho(t+dt) = (1-\delta p)|\psi_{\text{no jump}}\rangle\langle\psi_{\text{no jump}}| + \sum_c \delta p_c|\psi_{\text{jump}}^{(c)}\rangle\langle\psi_{\text{jump}}^{(c)}| \tag{3.47}$$

and upon substituting the two definitions 3.45 and 3.46 in the first line and substituting the definition of the effective hamiltonian 3.40 in the third line,

$$\rho(t+dt) = |\tilde{\psi}(t+dt)\rangle\langle|\tilde{\psi}(t+dt)| + \sum_c C_c|\psi(t)\rangle\langle\psi(t)|C_c^\dagger\ dt \tag{3.48}$$

$$= (1 - iH_{\text{eff}}dt)|\psi(t)\rangle\langle\psi(t)|(1 + iH_{\text{eff}}^\dagger) + \sum_c C_c\rho C_c^\dagger\ dt \tag{3.49}$$

$$= \rho(t) - iH_{\text{eff}}\rho dt + i\rho H_{\text{eff}}^\dagger dt + \mathcal{O}(dt^2) + \sum_c C_c\rho C_c^\dagger dt \tag{3.50}$$

$$= \rho(t) - i[H,\rho]dt - \frac{1}{2}\sum_c\{C_c^\dagger C_c,\rho\}dt + \sum_c C_c\rho C_c^\dagger dt \tag{3.51}$$

$$\Rightarrow \dot{\rho} = -i[H,\rho]dt + \sum_c\left(C_c\rho C_c^\dagger - \frac{1}{2}\{C_c^\dagger C_c,\rho\}\right) \tag{3.52}$$

which reproduces equation 3.35 as required. This means the MCWF is a valid framework for simulating YbF molecules in a laser beam. For non-pure initial states, the input wavefunctions must be drawn randomly from a statistical ensemble with classical probabilities.

Letter	Value	Name
N	1	Rotational quantum number
I	1/2	Nuclear spin quantum number [45]
B	7233.8271	Rotational Constant [46]
γ	-13.41679	Electron-spin rotational coupling constant [46]
b	141.79014	Isotropic spin-spin interaction [47]
c	85.4208	Anisotropic spin-spin interaction [47]
C_I	0.02038	Nuclear-spin rotational coupling constant [46]
g_S	2.0023193043622	Electron spin g-factor (Default CODATA value) [48]

Table 3.1: Input parameters for *pyLCP* to calculate the ground state $X^2\Sigma^+(v=0)$ energies For quantum numbers, see table 2.1.

Letter	Name	Value
J	1/2	Angular momentum quantum number
I	1/2	Nuclear spin quantum number [45]
P	+1	Parity
a	7.2	Frosch and Foley a parameter

Table 3.2: Input parameters for *pyLCP* to calculate the ground state $A^2\Pi_{1/2}(v=0)$ energies. For quantum numbers, see table 2.1.

3.3 Implementation

3.3.1 Hamiltonian

The full hyperfine hamiltonian in the lab frame is given by

$$H = H_0 - \boldsymbol{\mu}_B \cdot \boldsymbol{B}_0 - \boldsymbol{\mu}_E \cdot \boldsymbol{E}(t) \tag{3.53}$$

where H_0 is the static hamiltonian dependent on nuclear spin $\boldsymbol{I}$, angular momenta $\boldsymbol{J}, \boldsymbol{L}$, rotational momenta N, etc and their interactions. The external magnetic field $\boldsymbol{B}_0$ is assumed to be constant and it was measured as $B_0 = 0.08\hat{x} - 0.13\hat{y} + 0.1\hat{z}$ G. H_0 can be calculated using *pyLCP* package. To calculate the energies, it requires an input of several experimental parameters. These are displayed in tables 3.1 and 3.2.

The energies from *pyLCP* are given as the eigenvalues of $\hat{h}_m$ described in detail in section 3.2.1. Call them ϵ_X and ϵ_A for a specific ground and excited state hyperfine manifold respectively. The laser frequency to connect the two levels is then

$$\omega_l = E_A + \epsilon_A - (E_{X_v} - \epsilon_X) + \Delta \tag{3.54}$$

with some internal detuning (such as sidebands, doppler, etc.) Δ. With the detuning derived in section 3.2.1, the code uses

$$\delta = \omega_l - E_A + E_{X_v} = \epsilon_A - \epsilon_X + \Delta. \tag{3.55}$$

pyLCP also returns the dipole operator, equivalent to the definition established in equation 3.23,

$$\hat{P}_X \hat{d}_q \hat{P}_A = d_0 (-1)^q \hat{\Sigma}^\dagger_{-q} \tag{3.56}$$

and then correcting for the Frank-Condon factor, the code's `d_blk` is exactly

$$C_{q,v} = (-1)^q \sqrt{f_v} \hat{\Sigma}^\dagger_{-q}. \tag{3.57}$$

The factor of d_0 is absorbed into the definition of $\Omega = d_0 \varepsilon$ which is determined by relating the saturation intensity

$$I_{\text{sat}} = \frac{\pi h c \Gamma}{3\lambda^3} \tag{3.58}$$

with

$$\Omega = \Gamma\sqrt{s/2}, \qquad s = I/I_{\text{sat}}. \tag{3.59}$$

The normalisation $\sum_q \hat{\Sigma}_q^\dagger \hat{\Sigma}_q = \hat{P}_A$ is validated against in the code numerically. The beam block 3.34 is then evaluated by taking the full hamiltonian including $\boldsymbol{d}\cdot\boldsymbol{\varepsilon}$ with no electric field and no detuning phase and subtracting the static hamiltonian, leaving only $\hat{M}_l$. It is evaluated at the transverse centre of the beam ($y = 0, z = 0$) where the Gaussian envelope is unity. The envelope is reapplied during the integration as a scaling of Ω_l by $\sqrt{s/s_{\max}}$. The reference point $\boldsymbol{r}_0$ is stratified randomly over one wavelength.

3.3.2 Integration

To find the wavefunction at the next time step, equation 3.41 must be integrated, i.e.

$$\frac{d\psi}{dt} = f(t,\psi), \quad f(t,\psi) = -iH_{\text{eff}}|\psi(t)\rangle. \tag{3.60}$$

The 4th order iterative Runge-Kutta method [49] is employed due to its high accuracy, stability, and as it is non-norm preserving. For a step size dt, the function f_n at step n is

$$f_{n+1} = f_n + \frac{dt}{6}(k_1 + 2k_2 + 2k_3 + k_4), \tag{3.61}$$

where

$$k_1 = f\left(t_n, f_n\right), \tag{3.62}$$

$$k_2 = f\left(t_n + \frac{dt}{2}, f_n + k_1\frac{dt}{2}\right), \tag{3.63}$$

$$k_3 = f\left(t_n + \frac{dt}{2}, f_n + k_2\frac{dt}{2}\right), \tag{3.64}$$

$$k_4 = f\left(t_n + dt, f_n + dtk_3\right). \tag{3.65}$$

The remainder of the MCWF is then performed as described in section 3.2.2. The times of photon scattering are recorded as well as statistics about which manifolds molecules accumulate in.

3.3.3 Microwaves

Microwaves play an important role in the ultracold experiment. In detector A, they couple the $|X, N = 0, F = 1\rangle$ level to the $|X, F = 1^+\rangle$ level. Unlike for the laser beams, the exact microwave field is not known. Assuming the microwave absorbing foam fully suppresses standing-wave structure, the field is modelled as a smooth travelling wave with no spatial envelope. It is treated as spatially uniform, always on, and perfectly monochromatic. The transfer from $|X, F = 1^+\rangle$ back down to $|X, N = 0, F = 1\rangle$ only occurs through rabi flopping because both are ground states and spontaneous decays are heavily suppressed and hence neglected. This means, the Lindblad loss operator does not apply. However, the dynamics can be absorbed into the static hamiltonian $\hat{H}_{\text{mw}}$. *pyLCP* is designed to produce dynamics associated with laser beams driving spontaneously decaying transitions and thus cannot be used for the microwave coupling. Hence, the dipole operator has to be derived explicitly. By the Wigner-Eckart theorem, the dipole operator in Hund's case b coupled basis $|N, J, F, m_F\rangle$ can be decomposed as

$$\langle N'J'F'm'_F|\hat{d}_q|NJFm_F\rangle = (-1)^{(F'-m'_F)}\begin{pmatrix} F' & 1 & F \\ -m'_F & q & m\end{pmatrix}\langle N'J'F'||\hat{d}||NJF\rangle \tag{3.66}$$

where the 2×3 array is the 3j-symbol and the reduced matrix element can be further decomposed as

$$\langle N'F'F'||d||NJF\rangle = (-1)^{(J'+I+F+1)}\sqrt{(2F'+1)(2F+1)}\begin{Bmatrix} J' & F' & I \\ F & J & 1\end{Bmatrix}\langle N'J'||\hat{d}||NJ\rangle \tag{3.67}$$

$$\langle N'J'||\hat{d}||NJ\rangle = (-1)^{(N'+S+J+1)}\sqrt{(2J'+1)(2J+1)}\begin{Bmatrix} N' & J' & S \\ J & N & 1\end{Bmatrix}\langle N'||\hat{d}||N\rangle \tag{3.68}$$

where we have followed Hund's case (b) when coupling the quantum numbers (see section 2.1 for more details). We set $\langle N'||\hat{d}||N\rangle = 1$ since for the $N = 0 \to N' = 1$ transition it encodes no unaccounted

for selection rules and its constant factor is absorbed into Ω at a later time. $\hat{\boldsymbol{d}}$ is then rotated from the coupled basis into the energy eigenbasis of both manifolds using $U_{N0}^T \hat{\boldsymbol{d}} U_{N1}$.

The polarisation of the microwave is assumed to be linear and coherent, oriented along $\boldsymbol{\varepsilon} = \frac{1}{\sqrt{2}}(\hat{y} + \hat{z})$, i.e. at 45° to the beamline which is the orientation of the microwave horns. The raising operator up to a constant is

$$\hat{\boldsymbol{d}}^+ \cdot \boldsymbol{\varepsilon} \propto \hat{C}_{\text{mw}} = \sum_q (-1)^q \hat{\Sigma}_q \varepsilon_{-q}. \tag{3.69}$$

The constant of proportionality is calibrated against the desired rabi frequency Ω_{mw}. However, there is an ambiguity due to multiple off-diagonal couplings. The coupling between two degenerate manifolds is a rectangular matrix, so there is no single off-diagonal element to identify with Ω_{mw}. Decomposing $\hat{C}_{\text{mw}} = \sum_i \sigma_i |u_i\rangle\langle v_i|$ splits the problem into independent two-level systems. The i-th oscillates at $2\sigma_i$. The input parameter Ω_{mw} was chosen to obey the convention of being defined as the largest component, so $\hat{C}_{\text{mw}}$ is rescaled by $\Omega/2\sigma_i$.

We apply the RWA to the lower ground state with general unitary operator

$$W = \exp\left(it(\theta_1 \hat{P}_{N1} + \theta_0 \hat{P}_{N0})\right). \tag{3.70}$$

This has the requirement that the $N = 1$ block must be exactly $\hat{h}_{N1}$ because it is already in the $\theta_m = E_m$ frame. Hence, θ_1 is not a free parameter and must equal $\omega_{\text{rot}} = E_{N1} - E_{N0}$, the difference between the two manifolds' centres. To make the coupling static, we choose $\theta_0 = \omega_{\text{rot}} - \omega_{\text{mw}} = E_{N0,F1} - E_{N1,F1^+} + \Delta$ with detuning Δ, and the resulting hamiltonian in the RWA frame is

$$\tilde{H}_{\text{mw}} \approx \begin{pmatrix} \hat{h}_{N1} & \hat{C}_{\text{mw}} \\ \hat{C}_{\text{mw}}^\dagger & \tilde{h}_{N0} \end{pmatrix}, \quad \tilde{h}_{N0} = \hat{h}_{N0} - (E_{N1,F1^+} - E_{N0,F1} - \Delta)\hat{1}_{4\times 4} \tag{3.71}$$

where $\hat{h}_{N0}$ and $\hat{h}_{N1}$ are the hyperfine hamiltonians of the two manifolds and Δ is the microwave detuning which is set to 0 on resonance. $\tilde{H}_{\text{mw}}$ is absorbed into the static hamiltonian which drives the entire MCWF.

Finally, the molecules experience the microwave field from entering the detection chamber well before moving through the laser beam. As a result, for the $N = 0, F = 1 \to N = 1, F = 1^+$ transition, approximately 33% will be in the $N = 1$ manifold upon entering the laser beams due to dark states forming. This would be too computationally expensive to simulate with a full MCWF simulation for the entire transition period before the laser beams. Thus, they are pre-evolved using a unitary static hamiltonian

$$U(t) = e^{-i\tilde{H}_{\text{mw}}(t+\eta)}. \tag{3.72}$$

where η is a random jitter spanning multiple Rabi frequencies to ensure that molecules are not correlated.

3.3.4 Polarisation Modulation

The laser beams employed in the ultracold experiment use polarisation modulation at $\omega_{f_{\text{EOM}}} = 1$ MHz to remix dark states using EOMs as described in part 2.4.5. This is a significant contributor to increasing scattering rates. In the simulation, beams propagate along $\pm\hat{x}$, so both $\hat{y}$ and $\hat{z}$ are transverse components. Defining the basis vectors

$$\hat{e}_1 = \frac{1}{\sqrt{2}}(\hat{x} + \hat{y}), \quad \hat{e}_2 = \frac{1}{\sqrt{2}}(\hat{x} - \hat{y}), \tag{3.73}$$

then a rotating polarisation vector can be rewritten

$$\hat{\varepsilon}(t) = e^{-i\phi(t)/2} = \frac{1}{\sqrt{2}}\left(e^{-i\phi(t)/2}\hat{e}_1 + e^{i\phi(t)/2}\hat{e}_2\right). \tag{3.74}$$

EOM theory was discussed in detail in section 2.4.5. If the potential across the crystal introduces noise into the sinusoid, lag at the rising or falling edge, or does not drop to 0 momentarily, $\hat{\varepsilon}$ will not be modulated perfectly. In the simulation, this is reflected by introducing a parameter β and writing

$$\hat{\varepsilon}(t) = \cos\left(\frac{\pi}{4} + \beta\right) e^{-i\phi(t)/2} + \sin\left(\frac{\pi}{4} + \beta\right) e^{i\phi(t)/2} \tag{3.75}$$

where $\beta = 0$ reproduces the ideal case 3.74. Now, applying $\phi(t) = \omega_{f_{\mathrm{EOM}}} t + \phi_0$ modulates the laser light in the cycle: linear along $\hat{y} \to \sigma^+ \to$ linear along $\hat{z} \to \sigma^-$. This factor is then multiplied by $\epsilon_{-q}^{(l)}$ of the beam block M_l defined in equation 3.34.

3.3.5 Laser Linewidth

So far, it was assumed that lasers are perfectly temporally coherent. This is not the case, as they have a linewidth, which is the width of the spectral power density of its frequency. Spontaneous emission in the laser medium and thermal noise set a fundamental limit on the linewidth [50]. The phase ϕ of a mode drifts according to a random brownian walk which can be modelled using the Wiener process [51]. The jitter after a time-delay dt can be characterised by variance [52]

$$\sigma^2 = 2\pi dt \Delta\nu_c \tag{3.76}$$

where $\Delta\nu_c$ is the lorentzian linewidth. Now using the expression of brownian motion $\sigma^2 = 2Ddt$, allows us to determine the diffusivity $D = \pi\Delta\nu_c$. A jitter step in phase is then distributed

$$\delta\phi = \sqrt{2Ddt}\,\mathcal{N}(0,1) \tag{3.77}$$

which is added to the laser detuning $\delta \to \delta + \delta\phi$.

3.3.6 Detector Geometry and Transverse temperature

Molecules with high transverse speeds may be more likely to traverse the wings of the gaussian laser intensity profile and thus scatter fewer photons. However, the aperture at before detector A will limit the maximum transmitted transverse velocity. Hence, MCWF has to take the detector geometry into account.

The reaction vessel produces YbF molecules which are thermalised to its 3.8 K temperature and are transmitted through an aperture of 3 mm r_S. They are then transversely cooled to less than 100 mK. They travel $R_s = 3.8$ m to the detection volume which itself has an aperture of $R_{\mathrm{ap}} = 5$ mm. The distance between this aperture and the detection laser is $d = 37$ cm. For a molecule leaving the first aperture at radius $s \in r_S$ and entering the simulation at radius r, then the maximum velocity that is not cut off by the apertures is

$$v_{\max}(r) = \left(\frac{rR_s + sd}{d + R_s} \pm r_s\right)\frac{v_f}{R_s} \tag{3.78}$$

where v_f is the forward velocity. The transverse velocity is then drawn from the Boltzmann distribution

$$f(v) = \sqrt{\frac{m}{2\pi k_B T}} e^{-mv^2/2k_B T} \tag{3.79}$$

with $T = 100$ mK and respecting the upper and lower bounds.

3.4 Code Availability

The code as described above is available at `https://github.com/LukasWystemp/ultracold_eEDM_MCWF_simulation`.

4

Results and Discussion

4.1 Overview

The MCWF simulation propagates molecules at a given velocity through the detection volume. Its input parameters fall into two categories: implicit and explicit. Explicit parameters depend on the specific setup and run. They are displayed in table 4.1.

Symbol	Description	Default
P_{P1} / P_{V1}	P(1)/ V1 power	//
//	Number and position of sidebands and their power distribution	see 2.4.5
$\Delta\nu$	P(1)/ V1 linewidth	754 kHz for P(1), 541 kHz for V1
v	Forward velocity	170 ms^{-1}
L	Distance from detector laser to source	3.8 m
σ	Standard deviation of beam profile	2.45 mm
T	Transverse temperature	1.8 K
Ω	Microwave Rabi frequency	100 kHz
//	Microwave polarisation	45° to beamline
$\boldsymbol{B}$	Magnetic field	0.2 G
f_{EOM}	Polarisation modulation frequency	1 MHz
β	Effectiveness of the EOM to switch between polarisation states	0
δ	Laser and microwave detuning	0
t_{pre}	Whether molecules evolve in microwaves before entering the laser beams	yes

Table 4.1: MCWF Explicit Default Input Parameters

The implicit input parameters on the other hand describe the general detector and simulation layout, such as aperture sizes, time steps, simulation starting points, etc. These are embedded more deeply in the code and altering them requires some familiarity with it. The raw output of a run includes the average number of photons scattered per molecule in transit, the state population as a function of time, a simulated CCD image, photon statistics, as well as various interim results. An example is shown in figure 4.1.

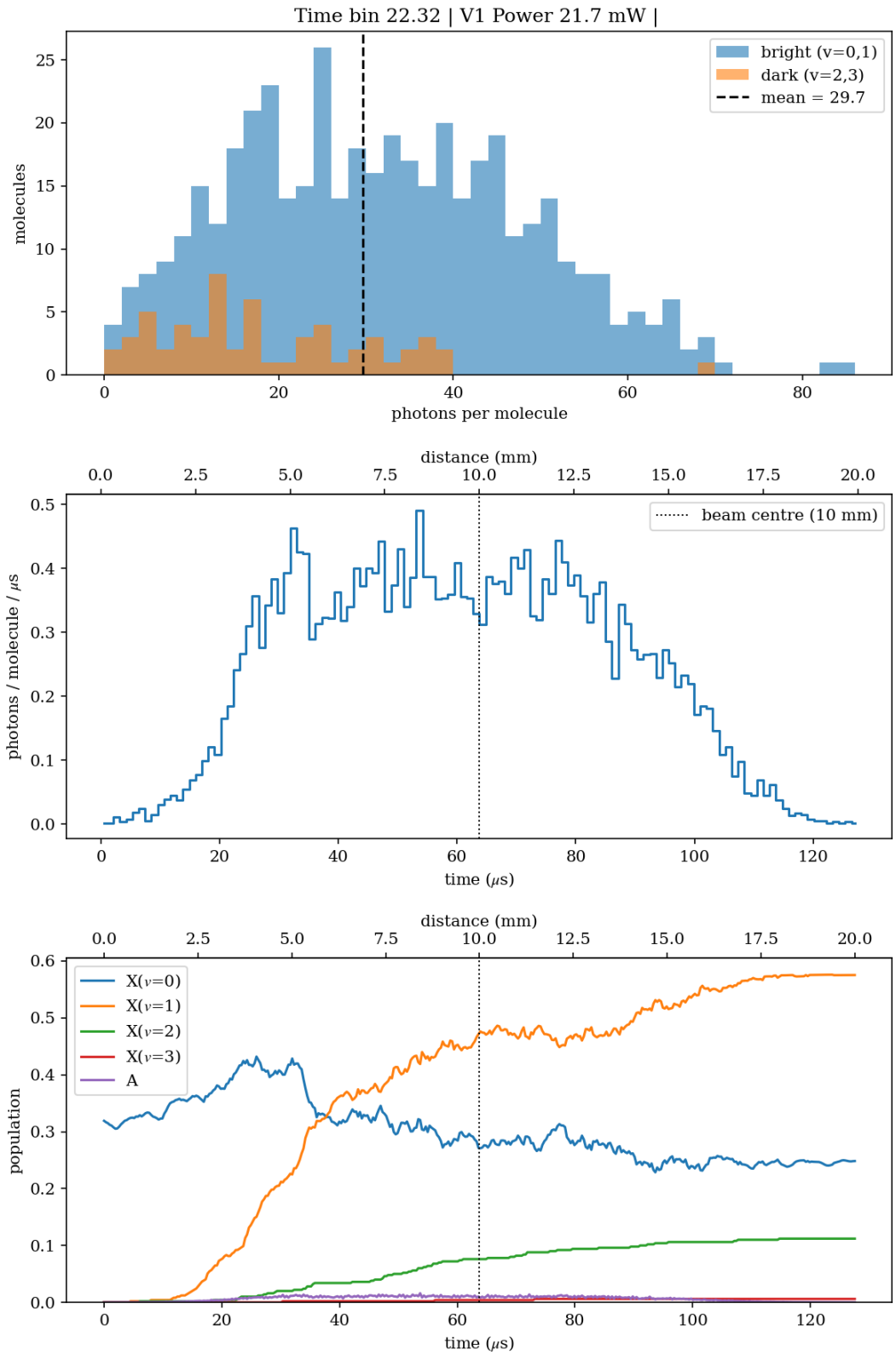


Figure 4.1: **Top**: Example Photon distribution **Middle**: Example CCD Image both on a time and distance scale. **Bottom**: Example fractional population distribution. Note, that it does not always add to 1 since some molecules are in $|X, N = 0\rangle$ and get continuously moved up my microwaves.

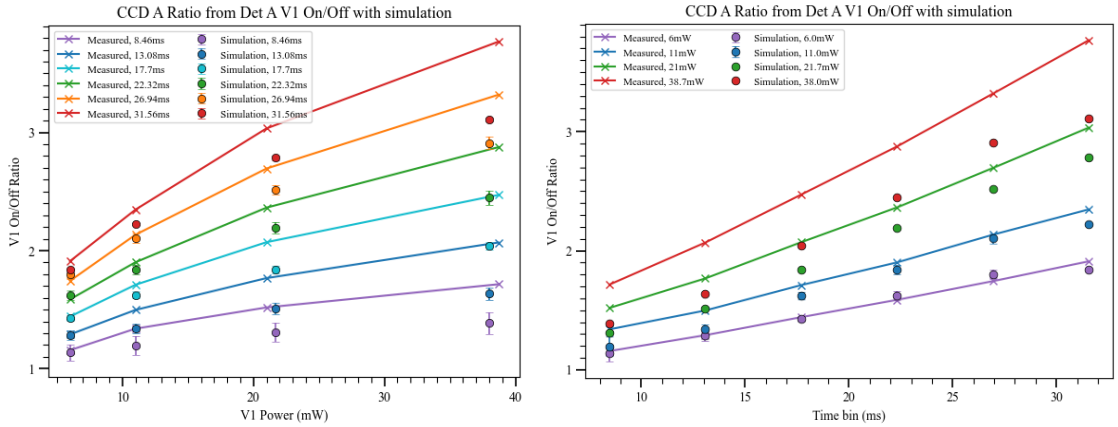


Figure 4.2: Simulation Results (dots) and Experimental $R_{\text{on/off}}$ (lines) with $P_{P1} = 105$ mW **Left**: V1 On/off ratio as a function of V1 Power **Right**: V1 On/Off ratio as a function of time bin

In the following sections, the MCWF simulation is validated against data with a scan over P1 powers and time bins, for different EOM polarisation modulation frequencies, and microwave powers. Then, a convergence test is performed to check the numerical stability of the simulation. It is compared to rate equations to extract the Rabi frequency from a previously measured rate. Then, it is utilised to inform B-field and modulation measurements to increase the V1 scattering enhancement.

4.2 Validation

4.2.1 Agreement and Disagreement from Experiment

The accuracy of the MCWF must be assessed across a range of applications. Because neither the absolute number of molecules nor their velocity distribution is known precisely, all comparisons are done in terms of ratios of one parameter to another. Working with ratios cancels many one-sided systematic uncertainties and removes any need to know the absolute molecule number. For the purpose of maximising the number of photons scattered in the detection region, the V1 On/Off ratio $R_{\text{on/off}}$, the number of photons detected with both the P(1) and V1 lasers on, divided by the number detected with V1 off, is a useful measure for the effectiveness of V1 light to improve detection efficiency. Figure 4.2 presents the same experimental data in two ways: On the left, $R_{\text{on/off}}$ for different time bins is shown to increase as V1 power is increases. The right figure shows the dependency of $R_{\text{on/off}}$ for a given P_{V1} on the time-bin. The data confirm that the V1 beam substantially enhances the signal. Molecules with slower velocities experience a larger enhancement because they spend more time in the laser beam. This time dependence also implies that the scattering does not run to completion, that is, that not all molecules have been pumped into inaccessible states which is confirmed by the population evolution plots produced by the MCWF simulation. The power dependence may have several origins, including an intensity-limited scattering rate or power broadening that enlarges the effective interaction volume.

The simulated results are in very good agreement across all time bins at low P_{V1}, but increasingly underpredict $R_{\text{on/off}}$ as P_{V1} rises although it continues to follow correct trends. This is a systematic effect that suppresses the scattering from V1 light proportionally more than P(1). A plausible cause could be additional dark-state remixing channels in the experiment which are not present in the simulation. At low V1 power, the re-pump scattering rate is intensity limited and dark states are not the bottleneck. However, if molecules at high V1 power are pumped into coherent dark states faster than they can be destabilised, dynamics are limited by the remixing rate. This hypothesis is supported by figure 4.3 which was taken at low P_{V1}. As expected, the simulation agrees well with the data in this low power regime for all time-bins. However, when the EOM is turned off it predicts too small a $R_{\text{on/off}}$. In that case, the remixing in the simulation is purely based on the static background B-field, phase diffusion, and motion through the standing wave. If any of these is less effective than assumed, the resulting V1-on scattering rate would be decreased. Switching the EOM polarisation modulation on in the simulation then masks this model's deficiency, since the modulation itself remixes states.

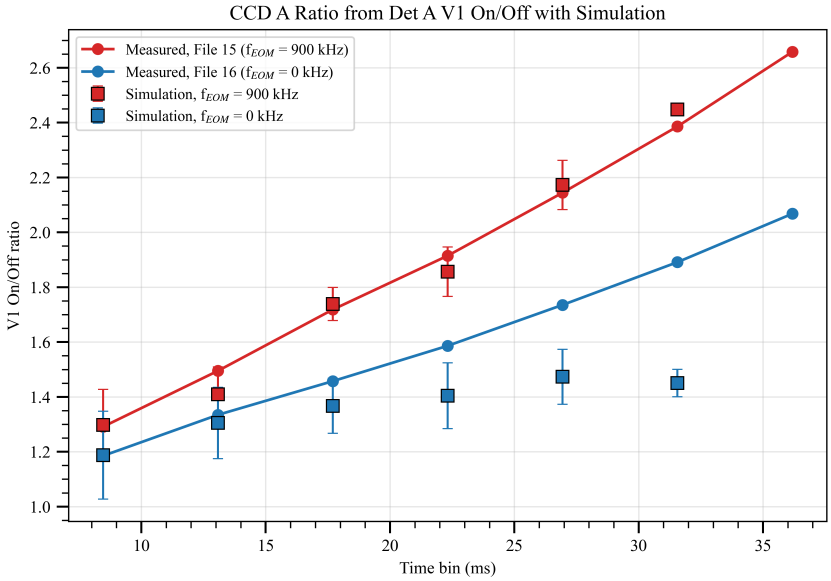


Figure 4.3: Measured and Simulated $R_{\text{on/off}}$ for $f_{\text{EOM}} = 900$ kHz and $f_{\text{EOM}} = 0$ with $P_{P1} = 105$ mW and $P_{V1} = 13$ mW.

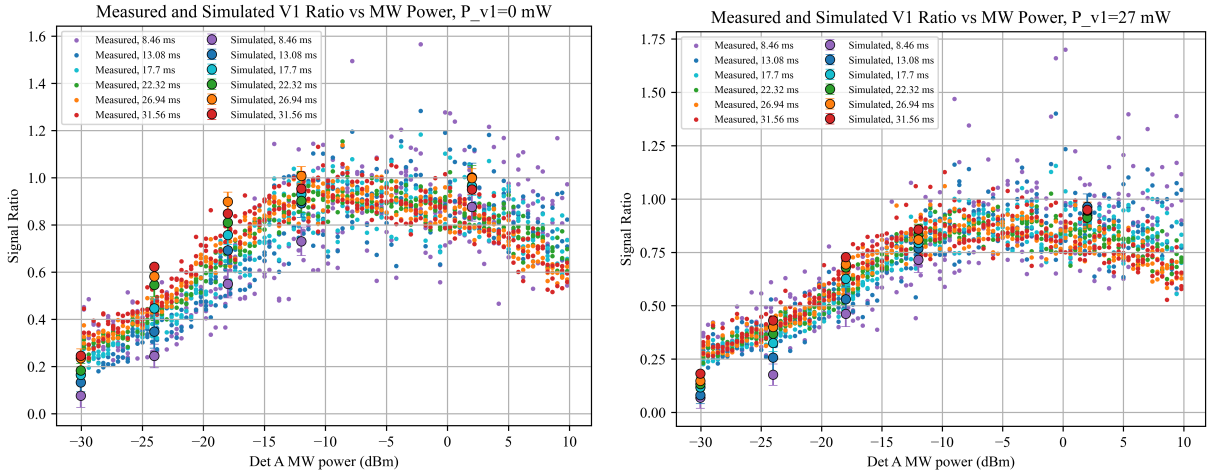


Figure 4.4: Microwave Power Scans with $P_{V1} = 0$ (left) and $P_{V1} = 27$ mW (right). $P_{P1} = 100$ mW.

Several candidates for the missing remixing exist. Currently, the model's polarisation modulation is purely temporal, whereas birefringence in the access ports and changes in polarisation of the returning beam from detector B in the opposite direction could both cause spatial polarisation gradients. These would remix dark states even when polarisation modulation is off. Additionally, the magnitude and direction of the B-field could change remixing dynamics. It was measured to the left and right of the detection chamber and interpolated between which brings high uncertainties. The non-magnetic stainless steel was assumed to be non-attenuating. However, a small change in the field could change mixing dynamics. This could be tested by sweeping the magnetic field direction in the simulation. Finally, the linewidth of the lasers is modelled as white phase diffusion perfectly correlated between all sidebands of a laser. However, frequency-dependent noise could effectively remix coherent dark states.

The third validation test concerns accuracy of the microwave coupling dynamics. Two power scans are shown in figure 4.4 for both V1 off (left) and at 27 mW (right). In both cases, the simulation closely matches the data. The discrepancy at high power above -5 dBM arises due to microwave leakage from the detection volume into the preceding P_{BG} Q(1) clean-up region. There, the microwaves move molecules

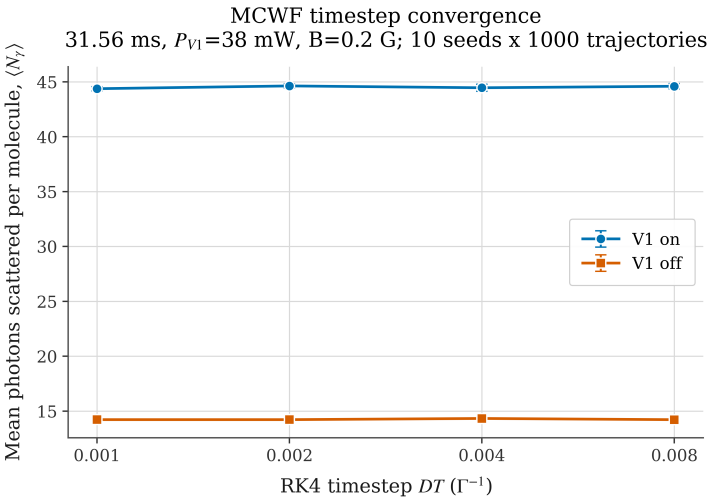


Figure 4.5: Convergence test for varying RK4 timesteps

from the measurement state $N = 0, F = 1$ into $N = 1$ where the Q(1) laser, intended to remove residual background, removes signal instead. The Q(1) laser is not included in this simulation and hence it does not reproduce this effect. The simulation also predicts too low a signal ratio at very low power below -25 dBm which is most likely due to residual background in $N = 1$, or in higher vibrational states which scatter even when microwave power is too low to re-populate the cycling states. It should be noted that the Rabi input frequency Ω_{mw} is related to the power in dBM by

$$P[\text{dBm}] = 20 \log(\Omega[\text{Hz}]) + 30 - K \tag{4.1}$$

where K is an empirical constant absorbing the coupling strength, attenuation in the cavity, scattering, and geometric factors. Here, it was found to be $K = 121$ dBm.

The investigation and comparison of the simulation to the V1 on/off ratio across V1 powers and time-bins, EOM frequencies, and microwave strengths together illustrate the power of the MCWF to describe complex detector interactions for many parameters. Measurements that would be complicated or time-consuming to take by hand can be approximated with this simulation.

4.2.2 Numerical Stability and Uncertainty

The MCWF simulation needs to be numerically stable to produce meaningful predictions. This means, across multiple runs, it needs to produce consistent results. This was validated by performing three simulations with $N_{MC} = 1000$ molecules each with different random seeds. These seeds produce distinct, uncorrelated sequences of random numbers which are used throughout the MCWF simulation. A simulation with a given seed is deterministic and produces identical results on every run, so varying the seed isolates the statistical spread. It was found that the standard deviation on the number of photons per molecule was $\sigma_{on} \approx 1$ with V1 on, and $\sigma_{off} \approx 0.5$ with V1 off. With the average number of scatters of the order of $\mathcal{O}(14)$ for V1 off and $\mathcal{O}(30)$ for V1 on, this amounts to no more than 4% uncertainty. Given the modest sample size of $N_{MC} = 1000$ molecules, the MCWF can be considered numerically stable.

A convergence test is crucial for establishing the appropriate time step size. In the correct regime, no observable should depend on it. The MCWF has restrictions for both small and large time steps. It must be small enough so that the norm decays approximately continuously satisfying $\delta p \ll 1$ at all times. At the same time, the Lindblad equation assumes a coarse-grain average of the density operator [42]. Hence, the time step needs to be larger than the coherence time of the vacuum. To see if this is the case, the MCWF simulation was run for a range of time steps in figure 4.5. No significant change in scattering dynamics is observed, confirming that the time step of 0.004 used for the remainder is a valid choice. This deserves further investigation as larger time steps can significantly speed up computation and it appears to only minimally impacts accuracy.

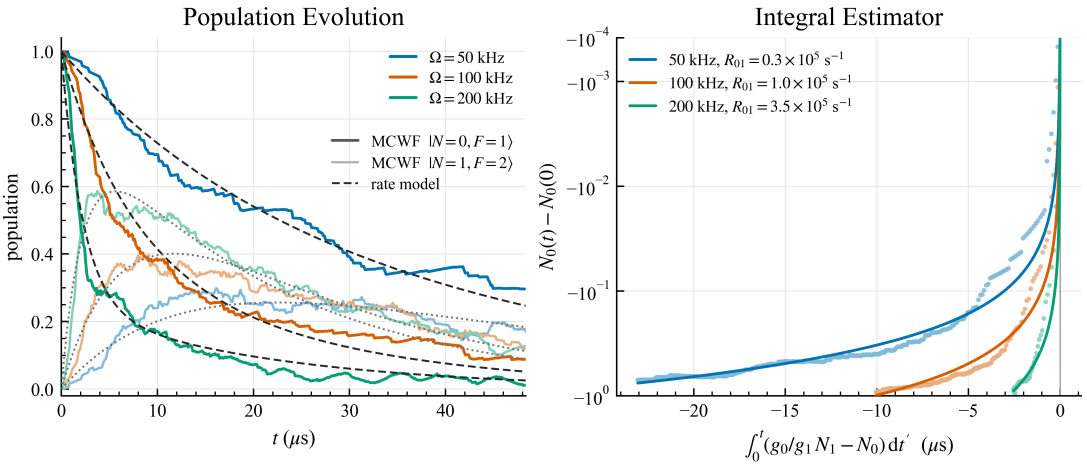


Figure 4.6: **Left**: Rate equation fit to populations of $|N = 0\rangle$ and $|N = 1\rangle$ predicted by the MCWF simulation for a selection of rabi rates **Right**: Integrated rate equation straight line with gradient R_{01} on a logarithmic scale

4.3 Microwave Investigation

Despite the failure of the current MCWF simulation to correctly predict dynamics at high P_{V1} power, it can accurately predict a wide range of results at low power. This allows us to investigate parameters which are difficult to measure experimentally.

Microwaves play a crucial role, coupling the measurement state into optically cyclable states. Previously, the signal ratio relative to the $N = 1$ population without optical pumping had been investigated using simple rate equations [14]. These are 4-state steady-state solutions where levels $|0\rangle$ (corresponding to $|X, N = 0, F = 1\rangle$), and $|1\rangle$ ($|X, N = 1, F = 2\rangle$) are coupled by microwaves at rate R_{01}. Note that in the version presented in this paper [14], the microwave scheme still operated under the old scheme of coupling to $F = 2$ instead of $F = 1^+$ as in the current scheme and the V1 laser was not part of the detection scheme. States $|1\rangle$ and $|e\rangle$ ($|A, J = 1/2, e\rangle$) are then coupled by a laser and they can decay back to $|1\rangle$ with ratio r or a dark state $|d\rangle$ with ratio $1 - r$. The MCWF simulation can give ready access to parameters which were previously inaccessible or challenging to measure, such as the rabi rate Ω_{mw}. This can be extracted by reproducing steady-state conditions where a stationary molecule is placed at the centre of the laser beam and evolved freely for a range of Ω_{mw}. The population evolution is shown in figure 4.6 left. Conversely, the rate model calculates populations with the simple coupled differential equations. For state $|0\rangle$ this is simply

$$\dot{N}_0 = R_{01}(N_1 - N_0). \tag{4.2}$$

Integrating and accounting for degeneracies $g_0 = 3$, $g_1 = 12$ gives

$$N_0(t) - N_0(0) = R_{01} \int_0^t \left(\frac{g_1}{g_0} N_1 - N_0 \right) dt' \tag{4.3}$$

which is a straight line graph through the origin with gradient R_{01} and plotted on a logarithmic scale on the right in figure 4.6. This allows us to relate the microwave scattering rate to the Rabi frequency. Given that the rate $R_{01} = 1 \times 10^5$ Hz at a power of 10 dBm is known [14], the corresponding Rabi frequency must be

$$\Omega_{\text{mw}} = 100 \text{ kHz}. \tag{4.4}$$

Figure 4.6 is a key result as it shows that the MCWF reduces to rate equations in the steady-state limit as expected. The small divergence from the rate equation model is due to its additional complexity where coherent effects are relevant.

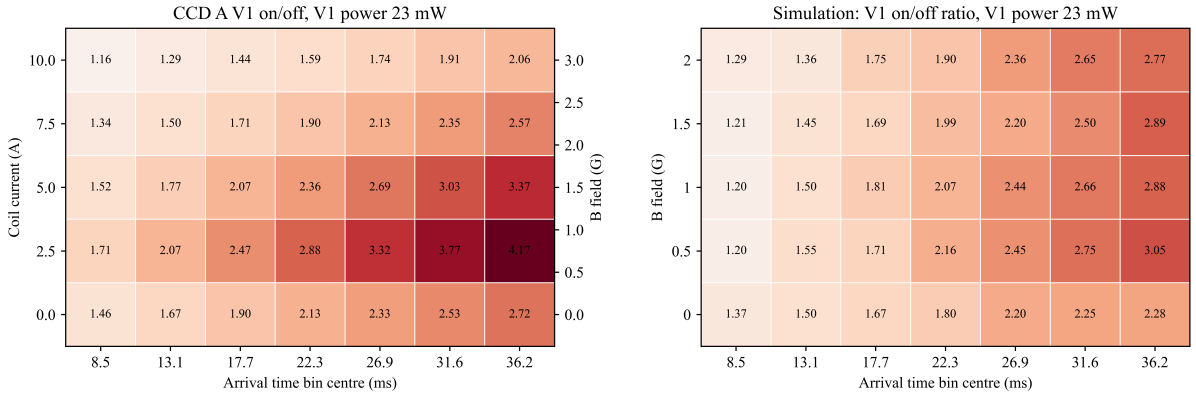


Figure 4.7: Heatmap of V1 On/Off Ratio as a function of Time bin (ms) and B-field. **Left**: Current with applied B-field **Right**: Simulated data with absolute B-field

4.4 Detection Improvements

The MCWF simulation is not merely descriptive, but can make predictions about the detection scheme. It was applied here to investigate how the V1 ratio enhancement could be improved further. Two parameters will be analysed in detail: The effects of B-fields and the EOM polarisation modulation.

4.4.1 B-field

Magnetic fields in the detection region have two competing effects. They remix dark states through Larmor precession which would enhance the scattering ratio, but strong magnetic fields detune the hyperfine levels through the Zeeman effect decreasing the microwave and laser coupling strengths. The net result was tested by applying a magnetic field using a Helmholtz-coil like setup at 45° to the beamline and the laser's $\boldsymbol{k}$-vector. At this angle, the remixing is optimal and hence this should serve as an upper bound. The magnetic field midway between each coil is then

$$B = \frac{8\mu_0 N I R^2}{(4R^2 + d^2)^{3/2}} \tag{4.5}$$

where N is the number of turns in each coil, R is the radius of both coils, and d is their separation, and I is the current. For this setup, $N = 20$, $R = 0.2$ m, such that in Gauss, $B = 0.3I$. The applied fields range from 0 to 3 G. Note that the background magnetic field in the experiment is around 0.2 G in addition to the applied B-field, whereas the simulation takes the total field as its input parameter.

The measured data is displayed in figure 4.7. A clear trend emerges where the V1 On/Off enhancement is largest between 0.5 and 1 G. The enhancement is largest at slow speeds (high time bins) which is expected as they spend more time in the laser beam and hence the B-field has more time to re-mix dark states. The simulation reproduces the general trend closely, but with some discrepancies. For one, the predicted ratio is consistently lower than observed. This is likely for similar reasons as discussed in part 4.2.1, where additional remixing channels exist which are not reflected in the MCWF simulation. the additional background fields in the experiment probably raise the measured scattering ratio further. 2 G, where the ratio is expected to fall but the simulation instead predicts a slight rise. This is most likely statistical, given the small number of molecules simulated, and it was not seen in other time bins.

In the experiment, the mean number of photons per molecule is a challenging result to derive directly as it is masked by the velocity distribution and the unknown absolute particle number. The MCWF provides it naturally and it is shown in figure 4.8 as a scan over B-field strength for different time bins. The mean scattering rate increases for both the V1-on and V1-off cases, though proportionally more when V1 is on. With the re-pump laser present, molecules scatter more before going dark, giving the field more opportunity to remix states. The peak lies at 0.5 G which is at least 0.3 G above the background field. This is a key result for deciding whether adding a field to the detection region can genuinely increase the signal, since a fall in the V1-off scattering number could have been hidden inside the ratio while in fact worsening the statistical uncertainty of the detector. Being able to separate such effects and give

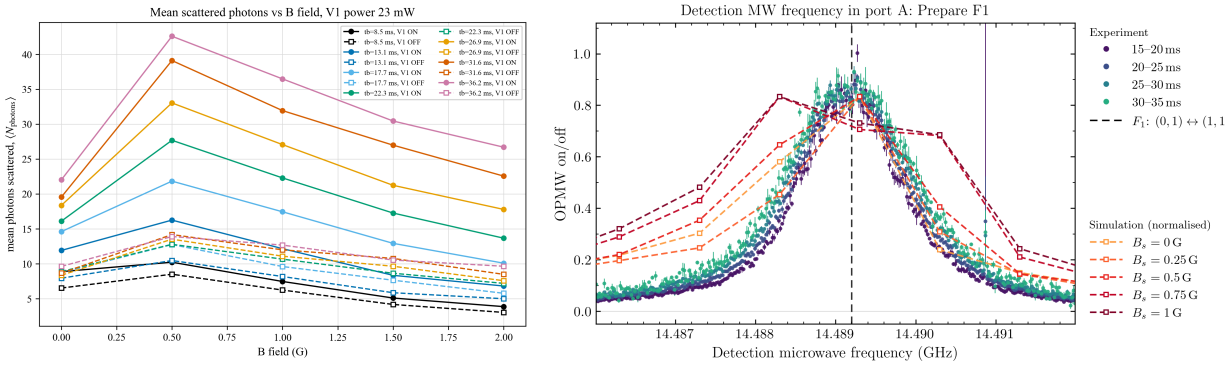


Figure 4.8: **Left**: Average number of photons scatterd per molecule for various time bins with V1 power at 23 mW and off. The statistical errorbars are too small to be visible. **Right**: Scattering ratio with optical pumping as a function of microwave detuning

fine-grained predictions is one of the key applications of the MCWF.

To isolate the effect of Zeeman-induced microwave detuning, the microwave detuning was scanned over in figure 4.8 right. The MCWF simulation results for various magnetic fields are overlaid. For 0.25 G, close to the background field, the MCWF follows the measured lineshape very closely with the correct amplitude and waist. The deviation at lower frequencies arises because the data were taken with the P(1) cycling laser at only 30 mW while the simulation was run at 100 mW, introducing additional power broadening. For higher B-fields, the waist becomes larger. This is due to Zeeman splitting which increases with larger B and causes more off-resonant driving. The peak falls markedly from 0.9 to 0.7. This confirms that while small B-fields (around 0.5 - 1G) can increase the scattering ratio and hence sensitivity of the detector through dark-state re-mixing, the Zeeman effect at larger B-fields leads to a decrease in coupling strength.

4.4.2 Polarisation Modulation

Having recognised that dark-state remixing is a significant lever on the scattering ratio, the effects of the frequency of the EOM polarisation modulation are investigated. However the EOM used in the ultracold EDM experiment is a custom-built resonant EOM, so its frequency cannot be adjusted. This is where the MCWF simulation serves as particularly useful in that it allows us to investigate this otherwise inaccessible parameter space. A simulated scan over P1 power is shown in figure 4.9. Even at low modulation frequencies, the EOM is extremely effective at increasing the scattering rate, over 2-times at low V1 power (≈ 6 mW). Raising the frequency further then only yields minimal gains. This is likely because the dark-state remixing rate is saturated agains the scattering rate. The scattering rate for the P(1) cycling laser is of the order of $\sim 10^6$ Hz [14]. When ω_{EOM} exceeds this, molecules are remixed before they can form dark states. At low V1 power, the pumping rate is small and so even modest polarisation modulation can reach the threshold. As the power increases, curves of different f_{EOM} fan out as low frequencies are no longer sufficient. In practice this means that the returns from increasing the modulation frequency diminish rapidly, even at high V1 power.

4.5 Discussion

4.5.1 Reliability of MCWF simulation

The simulation identified the correct trends in all eight scenarios tested above, demonstrating extraordinary predictive power. The numerical values, however, are often inaccurate deviating up to 40% deviation from experimental data, especially at high power. There are two reasons for this. First, some input parameters are poorly known and have to be estimated with high uncertainties. These include magnetic fields within the detection chamber, how switching effectiveness of the polarisation of the EOM, the transverse temperature, etc. The MCWF can be quite sensitive to changes in these parameters and their uncertainties propagate through to the output. Additionally, some parameters may not be included in

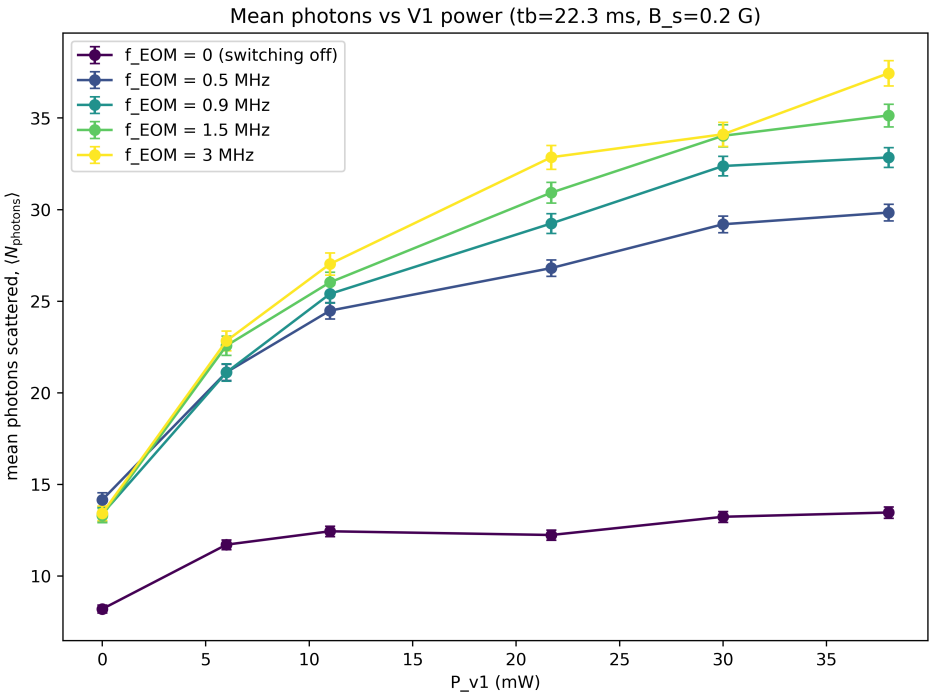


Figure 4.9: Number of photons scatterd per molecule for specific EOM frequencies as a function of V1 power with statistical uncertainty.

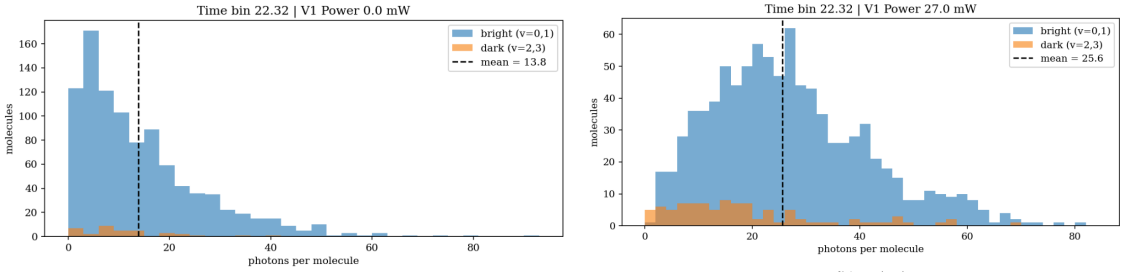


Figure 4.10: Number of photons scattered per molecule with $P_{P1} = 90$ mW **Left**: V1 off **Right**: V1 on at 21 mW

the model. For example, spatial polarisation gradients were identified to be a missing factor. Including these can greatly improve the accuracy of the model. However, every additional parameter also brings its own estimation uncertainty, which may offset the gain. This prompts an empirical approach in which data inform the MCWF simulation and the MCWF in turn indicates which parameters most warrant investigation. The direction of the B-field in the detection chamber is a good example: it is poorly known and may remix dark states more effectively than assumed, and sweeping it in the simulation would show how strongly the accuracy depends on it.

4.5.2 Capacity for Increasing the Detection Efficiency

Using the MCWF in combination with the detection improvements identified above allows the achievable reduction in noise to be calculated using the excess noise factor F introduced section 2.4.4. The MCWF simulation was run with standard parameters to obtain realistic photon statistics in figure 4.10. With V1 off, the distribution is approximately exponential $e^{-b_l x}$ with b_l the fraction of molecules going dark and is close to the Frank-Condon factor. On average, $\bar{n} = 13.8$ photons are scattered which is very close to the expected published value of 13.4 [14]. The standard deviation is estimated to be $\sigma_n = 11.34$. With V1 on, the mean increases to $\bar{n}' = 25.6$ and the standard deviation increases slightly to $\sigma_n' = 13.7$. The

standard formula for the excess noise factor [32] is

$$F = 1 + \frac{1}{\bar{n}}\left(\frac{1}{\epsilon} - 1\right) + \frac{\sigma_n^2}{\bar{n}^2} \tag{4.6}$$

where ϵ is the probability of detecting a photon including optical efficiency and quantum efficiency. Using the published result that on average, a shot includes 2×10^6 molecules and 10^6 photons are detected [14], this is estimated to be $\epsilon = 0.0362$. We therefore obtain

$$\sqrt{F} = 1.9 \rightarrow \sqrt{F'} = 1.5 \tag{4.7}$$

giving a improvement factor of 1.3. This should be taken as a lower bound because the MCWF underestimates the photon scattered at high V1 power and the result is therefore somewhat better. F is heavily limited by the photon shot limit $1/\bar{n}\epsilon$. Since the detection probability ϵ is set by both geometric factors and the efficiency of the CCD cameras, this parameter is difficult to optimise further. Instead, $\bar{n}$ can be improved. Raising the V1 power increases the scattering, but as figure 4.2 shows, the gains flatten quickly once the process becomes remixing limited. Indeed, the MCWF simulation found that increasing the V1 power to 38.5 mW raised the mean to 28 photons per molecule on average - a small improvement. An optimal B-field between 0.5 - 1 G could increase the V1 on/off ratio to 2.88 in the most central time-bin. Combining this with optimal polarisation modulation at 3 MHz, $\bar{n}$ could reach 40 photons per molecule as verified by the simulation. Under these optimal conditions, $\sqrt{F'}$ would drop to 1.34 making a sensitivity improvement of at least 1.4 possible.

Noise is not the only consideration, however. After scattering, the MCWF shows that around 50% of molecules are in the $|X, v = 1\rangle$ manifold while 30% are in $|X, v = 0, N = 1\rangle$. This would be a significant contributor to x_B noise raising the noise in detector B enough to drown the signal entirely. As such, this improvement hinges on whether a Q(1) clean-up laser can pump the remaining molecules into the $|X, v = 0, N = 3\rangle$ manifold. While this Q(1) laser was installed (see appendix B), assessing its effectiveness at cleaning up the remaining populations lies beyond the scope of this thesis.

5

Conclusion

In this thesis, a Monte-Carlo wavefunction simulation based on the Lindblad equation was implemented and validated against experimental data. It was found to be accurate at low power but diverge at high power due to missing dark-state re-mixing channels. Further uncertainties arise due to a lack of information regarding various input parameters, such as transverse temperature, effectiveness of the polarisation modulation, and others. Despite its large uncertainties, the MCWF simulation is highly numerically stable and could identify correct trends in the data. It finds significant applications within the ultracold EDM experiment as it allows us to investigate parameter space which would otherwise not be reachable, like the EOM polarisation modulation frequency. Furthermore, it has predictive power due to its comprehensiveness and can guide further research. It was employed extensively to investigate scattering processes in detector A. To improve the excess noise factor F and reduce statistical uncertainty, a V1 re-pump laser was added to pump molecules from the $v = 1$ vibrational state back and optically cycle for longer. It was found that a small magnetic field between 0.5 and 1 G can improve the V1 enhancement, however, a larger B-field detunes the microwave coupling leading to a strong reduction in coupling strength. EOM polarisation modulation at higher frequencies modulation was found to have a limited impact in comparison to lower frequencies with increasingly diminishing returns at higher V1 power. The new detection scheme could hence reduces the excess noise factor from $\sqrt{F} = 1.9$ to $\sqrt{F'} = 1.5$, reducing the statistical uncertainty of the experiment by a factor of 1.3 with potential to quickly obtain a reduction by 1.4 assuming optimal choice of B-field and polarisation modulation, and transversely cooled temperature. The ability to implement this scheme without increasing x_B noise significantly hinges on the effectiveness of a clean-up laser to remove the increased residual population in the $N = 1$ manifold.

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Appendix A

YbF Levelscheme

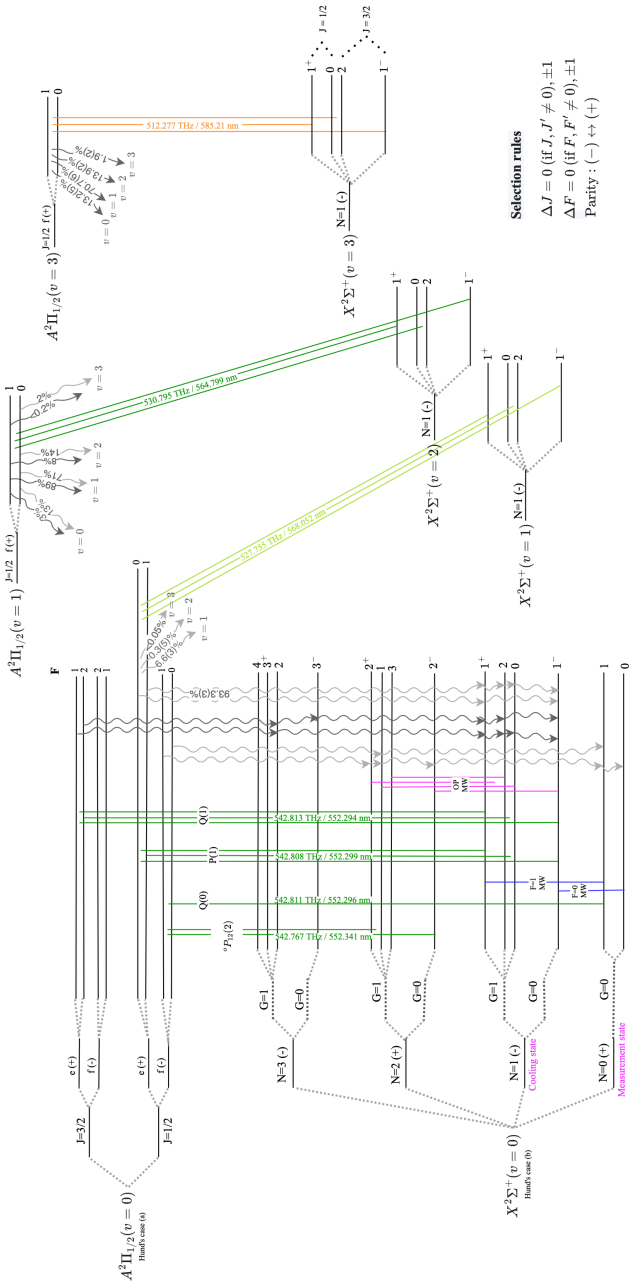


Figure A.1: Full YbF Energy-Level diagram used in the ultracold eEDM experiment with hyperfine components, laser transitions, microwave transitions, all decay channels, and selection rules.

Appendix B

Clean-up Extension Installation

To improve the detection signal-to-noise ratio (SNR), a V1 repump laser was added to the P(1) cycling laser beam as well as the clean-up. This is because with the V(1) laser added, molecules do not cycle to completion within the timeframe of traversing the beam, hence accumulating in the $N = 1$ manifold which would contribute to the x_B noise. Hence, an additional clean-up laser after the P(1) laser is required. This would pump molecules from $|X, v = 0, N = 1\rangle$ into $|A, J = 3/2, e\rangle$ from which has decay channels to $|X, v = 0, N = 3\rangle$ and to higher vibrational states. If a V1 repump laser is also added to detector B, then Q(1) clean-up also requires this V1 light so that the $|X, v = 1, N = 1\rangle$ manifold does not contribute to x_B. Within the scope of this project, the V1 light was combined with the Q1 light and coupled to the detector. The mirror-setup is shown in figure B.1 on the left. To combine the two wavelengths, a dichroic mirror was used whose transmission and reflection efficiency depends on the angle. Since both beams have to be overlapped at the fiber, the transmitted and reflected beam's angles are directly related by a shift which is plotted in figure B.1 right. This allows the dichroic mirror to be positioned such that the coupling is optimal.

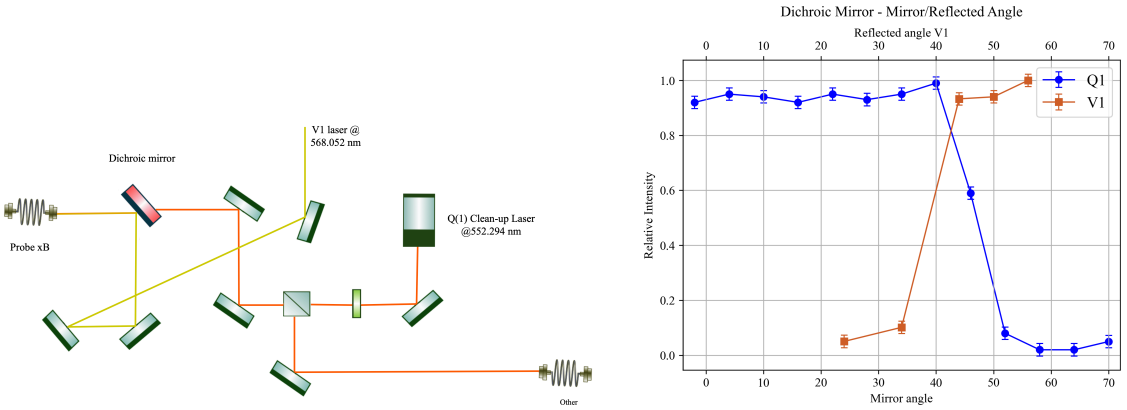


Figure B.1: **Left**: Mirror setup to couple V1 into Q(1) **Right**: Determining the optimal reflected and transmitted angle for the dichroic mirror